Neighbor Spectra Maintenance and Context Affinity Enhancement for Single Hyperspectral Image Super-Resolution

Heng Wang, Cong Wang, *Member, IEEE*, and Yuan Yuan, *Senior Member, IEEE*

***Abstract*— Single hyperspectral image super-resolution (HIS) aims to improve the spatial resolution of a hyperspectral image without relying on auxiliary information. By taking advantage of the high similarity among neighbor bands, some recent methods have used a recursive structure to super-resolve a hyperspectral image band-by-band. They are usually memory-efficient and perform well. However, they tend to introduce feedback information without distinction so as to weaken the utilization of complementary information in the context. In addition, the spectral structure is inevitably destroyed when spatial information is extracted from neighbor bands, which hampers the effective exploration of spectral information in the subsequent process. To this end, we propose a two-stage network based on neighbor spectra maintenance and context affinity enhancement (AE), which is composed of two subnetworks: neighbor network and context network. The former uses several neighbor bands to generate the neighbor spatial–spectral feature, incorporating a parallel processing scheme designed to reduce spectral distortion. Then we construct a relationship representation between the neighbor feature and feedback context information in the context network. By referring to the representation, the contents with higher complementarity will be highlighted in this stage. Experimental results on five public hyperspectral image datasets demonstrate that the proposed network not only outperforms state-of-the-art methods in terms of spatial reconstruction accuracy and spectral fidelity but also requires less memory usage.**

***Index Terms*— Affinity enhancement (AE), complementary information hyperspectral image, image super-resolution (SR), two-stage network.**

Manuscript received 10 August 2023; revised 9 February 2024; accepted 11 April 2024. Date of publication 16 April 2024; date of current version 25 April 2024. This work was supported in part by the National Science Fund for Distinguished Young Scholars under Grant 61825603, in part by the National Natural Science Foundation of China under Grant 62206220, in part by the Young Talent Fund of Association for Science and Technology in Shaanxi, China under Grant 20230140, in part by the Guangdong Basic and Applied Basic Research Foundation under Grant 2021A1515110019, in part by the Chunhui Program of Ministry of Education of China under Grant HZKY20220537, and in part by the Fundamental Research Funds for the Central Universities under Grant G2023KY0601. *(Corresponding author: Yuan Yuan.)*

Heng Wang is with the School of Computer Science and the School of Artificial Intelligence, OPtics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, China (e-mail: wangheng1@mail.nwpu.edu.cn).

Cong Wang is with the School of Artificial Intelligence, OPtics and ElectroNics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, China, and also with the Research & Development Institute of Northwestern Polytechnical University in Shenzhen, Shenzhen 518057, China (e-mail: congwang0705@nwpu.edu.cn).

Yuan Yuan is with the School of Artificial Intelligence, OPtics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, China (e-mail: y.yuan1.ieee@gmail.com).

Digital Object Identifier 10.1109/TGRS.2024.3389098

I. Introduction

HYPERSPECTRAL imaging obtains tens to hundreds of spectral bands in the target region to form a hyperspectral image, and a continuous spectral curve is constructed on each spatial pixel [1]. Compared with nature and multispectral images, hyperspectral ones reflect subtle spectral characteristics of objects. Therefore, they have been widely used in many applications such as mineral identification [2], plant monitoring [3], and geological exploring [4]. Nevertheless, due to the limitation of the number of photons captured by sensors, hyperspectral images always have a low resolution in a spatial domain, thus diminishing the subsequent analysis for some recognition areas, such as change detection [5], anomaly detection [6], [7], and object classification [8], [9].

To super-resolve the spatial size without other auxiliary images, several single hyperspectral image super-resolution (HIS) methods are developed to model the statistical distributions and use various mathematical techniques, such as sparse representation [10], [11], [12], [13], nonnegative matrix factorization [14], [15], [16], and self-similarity-based method [17], [18], to accomplish image reconstruction. Nevertheless, handcrafted priors can only reflect finite aspects of hyperspectral data, and they are insufficient to explore various hyperspectral data.

Recently, due to the remarkable representation ability, convolutional neural networks (CNNs) are introduced to implement single HIS reconstruction after they succeed in nature images [19], [20], [21], [22], [23], [24], [25], [26]. To improve the spatial resolution, the early models constructed by 2-D convolution focus on the information in a spatial domain, which neglects the spectral information [19], [20]. Then, with the aid of the 3-D convolution or varieties of hybrid convolution blocks, the consecutive information among the spectra can gain a full analysis, which significantly improves the reconstruction performance [22], [23], [24], [25], [26]. However, the 3-D convolution inevitably brings in a large number of parameters. Meanwhile, most of the previous methods take the whole hyperspectral data as an input, thus causing a large memory occupation [22], [23], [24], [25], [26]. In this case, the networks may fail to operate on devices with restricted memory capacity.

To improve their applicability, the existing networks consider reducing the memory usage in two aspects, i.e., model

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:29:33 UTC from IEEE Xplore. Restrictions apply.

lightweight and model architecture design [21], [22], [23], [24], [25], [26], [27], [28], [29]. The former commonly involves incorporating efficient blocks and strategies (such as separable 3-D convolution, weights sharing, and channel squeezing) to reduce the number of parameters or feature layers [21], [22], [23], [24], [25], [26]. It is worth noting that these techniques tend to weaken the presentation capacity of the network, thereby limiting its overall performance. In addition, compared with natural images, hyperspectral data exhibit a higher spectral dimension, thus posing a greater degree of complexity for super-resolution (SR) mapping.

Recently, by taking advantage of the high similarity among neighbor bands, a popular manner is to construct a recursive structure to super-resolve hyperspectral images band-by-band [27], [28], [29]. Such a technique only requires a limited number of adjacent bands as an input, enabling significant reductions in memory requirements. Moreover, contextual features are captured via feedback embedding, further enhancing the performance of the method. For SR reconstruction, which is an ill-posed problem, these methods can attain a considerable result due to a further constraint on the solution space. Along this insight, the performance of SR reconstruction for hyperspectral images depends on two issues: one is *how to fully explore spatial and spectral information among neighbor spectral bands*; the other is *how to effectively use the complementary information of the context*.

As for the first issue, during the process of spatial information mining, the spectral structure of the intermediate results should also be preserved for subsequent spectral information exploitation, which is meaningful for spectral reconstruction. The existing methods extract spatial–spectral features simultaneously, in which the reference band and its neighbor ones are usually merged at the beginning for subsequent processing. In this case, the spectral structure is inevitably destroyed during feature exploitation, which is not conducive to the subsequent spectral information extraction.

Regarding the second issue, the conventional approach involves combining the previously estimated feature, obtained from the previous band, with the current one for reconstruction. The combination is typically achieved through addition or concatenation operations. As shown in Fig. 1, these methods intend to compensate for the previous band information to the reference band with the uniform weight without any discrimination. However, the complementarity of different embedded information components relative to the reference band varies significantly. Thus, the uniform manner cannot effectively construct the relationship between the current band and its previous one, resulting in the adverse impact of irrelevant information on the reconstruction result.

Based on the previous analysis, this article proposes a novel two-stage network based on neighbor spectra maintenance and context affinity enhancement (AE) for single HIS, abbreviated as MESR, as shown in Fig. 2. The proposed network is based on a recurrent mechanism, which is memory-efficient and composed of two subnetworks: neighbor network and context network.

To effectively exploit information among the neighbor bands, a parallel processing strategy is used in the neighbor network. On one hand, this manner avoids the mixing of neighborhood band information, and the content independence of the reference band can be maintained. Meanwhile, the potential complementary information in the neighborhood band can be explored under the guidance of the reference band. On the other hand, the spectral structure can be effectively preserved, which benefits subsequent extraction of spectral characteristic.

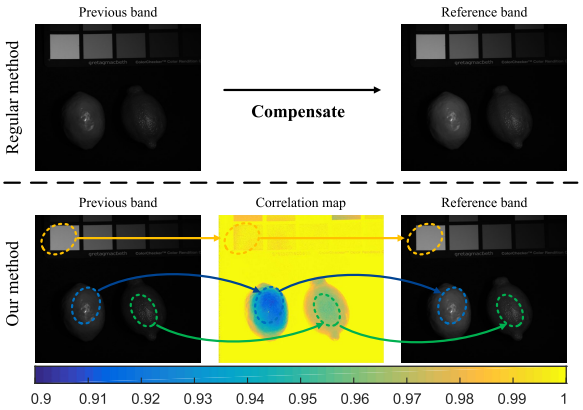


Fig. 1. Different ways of compensating the information within the previous band to the current one. The top row represents the regular manner, which treats all the complementary pixels in a uniform manner. The bottom row is our view, which constructs a correlation representation between the previous band and the current one and enhances the areas with more complementary information according to the representation map.

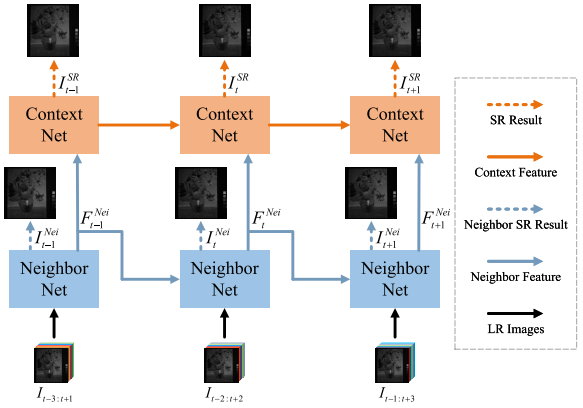


Fig. 2. Architecture of our whole network.

To construct the relationship between the current band and feedback feature, an AE module is designed to learn the correlation representation between them. This approach makes the context network pay more attention to the higher correlated content of feedback information from the previous band at the feature level, thereby enhancing the ability to capture more complementary information in the context.

In summary, the proposed network includes three advantages.

1) A parallel processing strategy is applied in the neighbor network to preserve the spectral structure during feature extraction, which is conducive to the subsequent exploration of spatial–spectral consistency information.
2) A cascaded AE module is developed in the context network to learn the difference representation between feedback information and current band information, which enhances the focus on strongly complementary content in feedback information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:29:33 UTC from IEEE Xplore. Restrictions apply.

3) By integrating the neighbor and context networks, a recursive framework for HIS is constructed, which achieves remarkable performance and high memory efficiency.

This article is organized as follows. Section II reviews some related studies of image SR. Section III introduces the proposed model in detail. Section IV presents the experimental results. Finally, Section V concludes this article.

II. Related Work

In this section, we briefly review the most related works of our proposed method, including the nature image SR and the single HIS. As our method is based on the CNNs, we present the deep-learning-based SR works here.

A. Single Nature Image SR

As is generally acknowledged, single nature image SR is the basis of the SR reconstruction area, from which single HIS has benefitted a lot. Dong et al. [31] first propose an SR convolutional neural network (SRCNN) to super-resolve the low-resolution (LR) image, which showed a better performance than the traditional algorithms. After that, taking advantage of the residual learning and dense connections, many CNN-based methods with deeper layers have been developed to improve the reconstruction results, such as the very deep network for SR (VDSR) [32], dense connected network (SRDenseNet) [33], and enhanced deep residual network (EDSR) [34]. Furthermore, DRCN [35], DRRN [36], and EBRN [37] introduce a recursive structure to increase the network depths without adding new parameters. For instance, Jiang et al. [38] propose a hierarchical dense recursive network (HDN) that incorporates abundant interleaved diagonal connections to fully extract and fuse features. Jiang et al. [39] develop a deep distillation recursive network (DDRN), in which a distillation and compensation strategy is constructed to address the high-frequency details lost during information propagation. Focusing on recovering finer texture details, Ledig et al. [40] introduce a generative adversarial network to super-resolve images. Recently, transformer-based architectures are increasingly prevalent, such as SwinIR [41] and GCnet [42], which can explore long-range dependency. Furthermore, Lu et al. [43] propose an efficient SR transformer (ESRT) for natural image SR, which has low computational cost and high memory efficiency. Besides constructing more powerful network architecture, many effective strategies are applied to improve their modeling ability and constantly refresh the state-of-the-art, including the attention mechanism [44], self-supervised training [45], and perceptual loss [47]. For example, Xiao et al. [45] propose a self-supervised degradation-guided adaptive network for blind SR, which aims to alleviate the domain gap between the simulated and real-world scenarios.

Although great achievements have been made, the above methods are not applicable to the HIS for two reasons. For one thing, these methods concentrate on the feature excavation in a spatial domain and neglect the consecutive information in a spectral domain, causing spectral distortion. For another, they lack consideration of the complementarity among the spectral bands, leading to incomplete information exploration.

B. Single Hyperspectral Image Super-Resolution

Recently, in the field of HIS, fusion-based techniques have been developed and achieved considerable performance [48], [49]. They enhance the spatial resolution of hyperspectral images by fusing with HR auxiliary information, such as panchromatic or red-green-blue images. Compared with these techniques, single HIS aims to improve the spatial resolution of an LR image without additional information, which is more practical and has aroused extensive attention. Yuan et al. [50] use a CNN to construct a mapping between LR and HR hyperspectral images with the idea of transfer learning and adopt collaborative nonnegative matrix factorization (CNMF) to constrain the result. Similarly, Xie et al. [14] super-resolve the selected bands with CNNs and integrate the complementary features by the nonnegative matrix factorization (NMF) method. However, these two-step methods are unstable and inefficient for the reconstruction of hyperspectral images. Without preprocessing or postprocessing, various end-to-end methods, adopting 2-D convolution, are designed to explore intrinsic property within the image [20], [52], [53]. For example, Li et al. [52] present a deep recursive residual network (GDRRN), where a groupwise convolution is introduced to reduce parameters. Liu et al. [53] develop an efficient unfolding network (EUNet) that incorporates the structure prior of hyperspectral images. However, these 2-D-based methods focus more on the reconstruction of spatial information and lack information exploration in the spectral domain.

To further explore the correlations of spatial and spectral information, 3-D convolutions are gradually applied [22], [23], [24], [30]. Mei et al. [30] first put forward a 3-D fully CNN (3DFCNN) to simultaneously exploit information in the spatial and spectral domains. Later, a variety of mixed 2-D and 3-D convolution structures are designed to extract effective complementary features. Li et al. [22] consider that spatial information is essential to the result while spectral features can be extracted, so they propose a mixed convolutional network (MCNet). Li et al. [24] explore the relationship between 2-D/3-D convolution for HIS (ERCSR), which can reduce the structure redundancy by sharing spatial information of the intermediate results. Capitalizing on the long-range modeling capabilities of transformers, Yang et al. [54] design a transformer-based framework (TC-HISRNet) to fully explore contextual information of adjacent spectra. Although these 3-D-based methods present considerable performance, they dramatically increase the number of network parameters and memory usage. In addition, these 3-D-based methods super-resolve all the spectra in parallel, and they have an advantage in information fusion along the spectrum. However, they suffer from a large model size and intensive memory consumption.

Recursive mechanisms could reconstruct the sequence data frame by frame, which have a small model size and are widely applied in video processing areas [55], [56], [57], [58]. Recently, inspired a lot by these and taking advantage of the high similarity between the neighbor bands, some researchers

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:29:33 UTC from IEEE Xplore. Restrictions apply.

take the spectral bands as frames in the video stream and super-resolve the hyperspectral data band-by-band [27], [28], [29]. For example, Wang et al. [28] propose a network using spectrum and feature context (SFCSR), where only neighbor bands are used as input to reduce the GPU memory budget. Wang et al. [27] design a recursive network to explore the complementary and consecutive information among the spectral bands and append a regularization network to preserve spatial–spectral coherence. To fully explore complementary information, Fu et al. [29] develop a bidirectional 3-D quasi-recurrent neural network to exploit global correlation along spectra. These recurrent networks combine context information through the feedback embedding from previous bands. However, all of them focus on how to extract useful features of the bands in each recurrent unit, while neglect how to effectively combine the feedback information, which increases information redundancy and is not conducive to information complementarity.

To this end, we design a two-stage network to construct the relationship between the current band and its feedback information, which can suppress the introduction of irrelevant information to improve the complementarity of context information. Meanwhile, to ensure the performance of the network, we also carefully design the spatial–spectral feature extraction for neighbor bands. Benefiting from the recursive mechanism, the proposed network can maintain good performance while requiring less memory footprint.

III. Proposed Methodology

In this section, to clearly lay out the proposed method, we first overview this two-stage method and then detail its two components: the neighbor network and the context network.

A. Overview

Given an LR hyperspectral image $\boldsymbol{H}^{\mathrm{lr}} \in \mathbb{R}^{w\times h\times L}$, where w, h are the width and height of the image, respectively, and L represents the number of spectral bands. The goal of the single HIS is to reconstruct the corresponding HR image $\boldsymbol{H}^{\mathrm{sr}} \in \mathbb{R}^{\alpha w\times \alpha h\times L}$ without auxiliary information, where α is the scale factor. For this purpose, considering the memory budgets in practical, we propose a recursive two-stage network.

For the input $\boldsymbol{H}^{\mathrm{lr}}$, let $\boldsymbol{I}_t^{\mathrm{lr}}$ be the tth spectral band, we split the spectral bands into L groups, i.e., $[\boldsymbol{h}_1^{\mathrm{lr}}, \ldots, \boldsymbol{h}_t^{\mathrm{lr}}, \ldots, \boldsymbol{h}_L^{\mathrm{lr}}]$. Each group contains the current band and its four neighbor bands, i.e.,

$$\boldsymbol{h}_t^{\mathrm{lr}} = \left[\boldsymbol{I}_{t-2}^{\mathrm{lr}}, \boldsymbol{I}_{t-1}^{\mathrm{lr}}, \boldsymbol{I}_t^{\mathrm{lr}}, \boldsymbol{I}_{t+1}^{\mathrm{lr}}, \boldsymbol{I}_{t+2}^{\mathrm{lr}}\right], 3 \le t \le L-2. \quad (1)$$

For the boundary bands, including $\boldsymbol{I}_1^{\mathrm{lr}}, \boldsymbol{I}_2^{\mathrm{lr}}, \boldsymbol{I}_{L-1}^{\mathrm{lr}}$, and $\boldsymbol{I}_L^{\mathrm{lr}}$, a periodic extension method is adopted here to form the corresponding groups. For example, $\boldsymbol{h}_2^{\mathrm{lr}}$ is expressed as $[\boldsymbol{I}_2^{\mathrm{lr}}, \boldsymbol{I}_1^{\mathrm{lr}}, \boldsymbol{I}_2^{\mathrm{lr}}, \boldsymbol{I}_3^{\mathrm{lr}}, \boldsymbol{I}_4^{\mathrm{lr}}]$. As illustrated in Fig. 2, these groups are then sequentially fed into the proposed recurrent network, and the HIS result can be obtained band-by-band.

Suppose that $\boldsymbol{I}_t^{\mathrm{lr}}$ $(3 \le t \le L-2)$ is the reference band, it can be super-resolved by processing its neighbor group $\boldsymbol{h}_t^{\mathrm{lr}}$. More specifically, our network consists of two subnetworks: the neighbor network and the context network. The spectral group $\boldsymbol{h}_t^{\mathrm{lr}}$, i.e., $\boldsymbol{I}_{t-2}^{\mathrm{lr}}, \boldsymbol{I}_{t-1}^{\mathrm{lr}}, \boldsymbol{I}_t^{\mathrm{lr}}, \boldsymbol{I}_{t+1}^{\mathrm{lr}}$, and $\boldsymbol{I}_{t+2}^{\mathrm{lr}}$, is first fed into the neighbor network to fully explore spatial and spectral information, producing a neighbor feature $\boldsymbol{F}_t^{\mathrm{Nei}}$ and a neighbor SR band $\boldsymbol{I}_t^{\mathrm{Nei}}$. Meanwhile, considering promoting the reliability of the intermediate results, the neighbor network also merges the consecutive information $\boldsymbol{F}_{t-1}^{\mathrm{Nei}}$ from the previous band via feedback embedding. This is formulated as

$$\boldsymbol{I}_t^{\mathrm{Nei}}, \boldsymbol{F}_t^{\mathrm{Nei}} = \mathrm{Net}_{\mathrm{nei}}\left(\boldsymbol{h}_t^{\mathrm{lr}}, \boldsymbol{F}_{t-1}^{\mathrm{Nei}}\right) \quad (2)$$

where $\mathrm{Net}_{\mathrm{nei}}$ is the neighbor network.

To effectively use the complementary information of the context $\boldsymbol{F}_{t-1}$, the context network constructs a relationship between the current band and the previous one at feature level, which can make it focus on the content with higher correlation. During this stage, the network outputs the SR reconstruction feature $\boldsymbol{F}_t$ and the final super-resolved band $\boldsymbol{I}_t^{\mathrm{SR}}$ by producing the neighbor feature $\boldsymbol{F}_{t-1}^{\mathrm{Nei}}$ and the context feedback $\boldsymbol{F}_{t-1}$, which can be expressed as

$$\boldsymbol{I}_t^{\mathrm{sr}}, \boldsymbol{F}_t = \mathrm{Net}_{\mathrm{con}}\left(\boldsymbol{F}_{t-1}, \boldsymbol{F}_t^{\mathrm{Nei}}\right) \quad (3)$$

where $\mathrm{Net}_{\mathrm{con}}$ is the context network.

After reconstructing all the HR spectral bands, we merge them together to make up the final HR hyperspectral image $\boldsymbol{H}^{\mathrm{sr}} = [\boldsymbol{I}_1^{\mathrm{sr}}, \ldots, \boldsymbol{I}_t^{\mathrm{sr}}, \ldots, \boldsymbol{I}_L^{\mathrm{sr}}]$.

B. Neighbor Network

Hyperspectral images have a remarkable characteristic that high similarity exists among the neighbor spectral bands. By taking advantage of this property, some researchers propose various recursive networks to super-resolve a hyperspectral image. Due to the high dimension of hyperspectral data, parallel processing methods suffer from intensive memory budgets. Compared with them, recursive methods take several bands as input to reduce the memory footprint. However, the existing methods focus on the extraction of spatial and spectral information, ignoring the spectral structure preservation on the intermediate results, which hinders the exploration for spatial–spectral consistency [59]. Recently, several temporal grouping strategies have been proposed in the field of video SR, which typically use reference frames as guidance to enhance the fusion efficiency of spatial–temporal information [56], [57], [58]. To this end, we introduce a band grouping strategy in the neighbor network to fully explore spatial and spectral information, as shown in Fig. 3.

Regarding $\boldsymbol{I}_t^{\mathrm{lr}}$ as the current band, we first divide the input spectral group $\boldsymbol{h}_t^{\mathrm{lr}} = [\boldsymbol{I}_{t-2}^{\mathrm{lr}}, \boldsymbol{I}_{t-1}^{\mathrm{lr}}, \boldsymbol{I}_t^{\mathrm{lr}}, \boldsymbol{I}_{t+1}^{\mathrm{lr}}, \boldsymbol{I}_{t+2}^{\mathrm{lr}}]$ into three subgroups according to their order relative to $\boldsymbol{I}_t^{\mathrm{lr}}$, i.e.,

$$\mathcal{G}\left(\boldsymbol{h}_t^{\mathrm{lr}}\right) = \left\{\left[\boldsymbol{I}_{t-2}^{\mathrm{lr}}, \boldsymbol{I}_{t-1}^{\mathrm{lr}}\right], \boldsymbol{I}_t^{\mathrm{lr}}, \left[\boldsymbol{I}_{t+1}^{\mathrm{lr}}, \boldsymbol{I}_{t+2}^{\mathrm{lr}}\right]\right\} \quad (4)$$

where $\mathcal{G}$ represents the group strategy. Note that the bands on the left and right appear in different subgroups. Passing through the 2-D convolution, the shallow features $(\boldsymbol{F}_0^L, \boldsymbol{F}_0, \boldsymbol{F}_0^R)$ corresponding to these spectrum subgroups are acquired. To improve the spatial resolution, the shallow feature $\boldsymbol{F}_0$ of the LR current band $\boldsymbol{I}_t^{\mathrm{lr}}$ is entered into several 2-D convolution blocks (2DBlocks) to explore its spatial information. Each 2Dblock is constructed by two residual (Res) modules

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:29:33 UTC from IEEE Xplore. Restrictions apply.

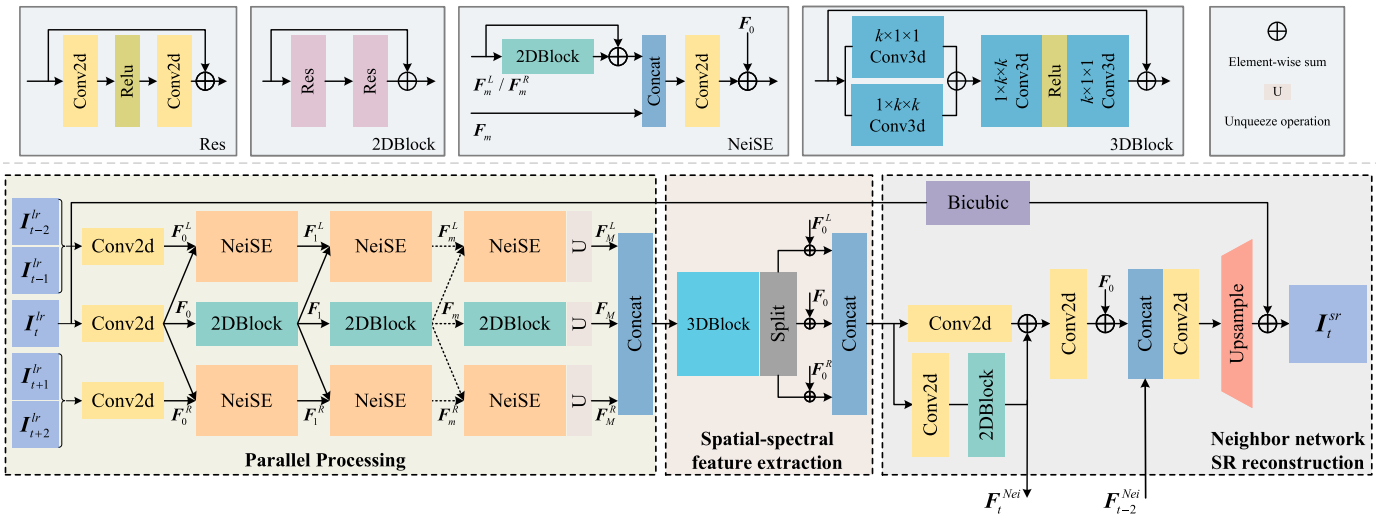


Fig. 3. Architecture of the proposed neighbor network.

and a skip connection, and the Res module is made up of 2-D convolutions. After passing through M 2DBlocks, we obtain the spatial feature $\boldsymbol{F}_M$ of $\boldsymbol{I}_t^{\text{lr}}$, which can be expressed as

$$\boldsymbol{F}_M = X_M(X_{M-1}(\ldots X_1(\boldsymbol{F}_0)\ldots)) \tag{5}$$

where X_m is the mth 2DBlock.

To fully explore the content among neighbor bands, the features of the left and right neighbor bands $(\boldsymbol{F}_0^L, \boldsymbol{F}_0^R)$ pass through several neighborhood spectra exploration (NeiSE) modules to acquire complementary information (see the upper and lower branches in Fig. 3). Note that the current band feature $\boldsymbol{F}_m$ is also introduced into the NeiSE module, which can improve the ability to extract complementary features from the neighbor bands. For the left and right neighbor bands $[\boldsymbol{I}_{t-2}^{\text{lr}}, \boldsymbol{I}_{t-1}^{\text{lr}}]$ and $[\boldsymbol{I}_{t+1}^{\text{lr}}, \boldsymbol{I}_{t+2}^{\text{lr}}]$, the complementary features $\boldsymbol{F}_M^L, \boldsymbol{F}_M^R$ from them can be obtained by

$$\boldsymbol{F}_M^L = Y_M\left(Y_{M-1}\left(\ldots Y_1\left(\boldsymbol{F}_0, \boldsymbol{F}_0^L\right)\ldots\right), \boldsymbol{F}_{M-1}^L\right) \tag{6}$$

and

$$\boldsymbol{F}_M^R = Y_M\left(Y_{M-1}\left(\ldots Y_1\left(\boldsymbol{F}_0, \boldsymbol{F}_0^R\right)\ldots\right), \boldsymbol{F}_{M-1}^R\right) \tag{7}$$

where Y_m is the mth NeiSE module, and there are M modules in total.

We fuse the neighbor features $\boldsymbol{F}_M^L, \boldsymbol{F}_M^R$ with the current one $\boldsymbol{F}_M$ by a concatenation way. In this parallel structure, each branch contains information of its respective subgroup. Since the spectral subgroups are divided according to the position relative to the current band, the spectral order is maintained at the feature level. Therefore, the spectral structure can be preserved to facilitate the subsequent exploration of spectral information.

The regular 3-D convolution would lead to a significant increase in network parameters and memory budgets. Inspired by the effectiveness and efficiency of separable 3-D convolution in processing temporal–spatial information, we design a 3-D convolution block (3DBlock) to explore spatial–spectral consistency information. Through the split operation, we adjust the size of the feature map for subsequent 2-D convolution operations. Before performing fusion, we also introduce the shallow features $\boldsymbol{F}_0^L, \boldsymbol{F}_0, \boldsymbol{F}_0^R$ into their respective branches, allowing the spectra structure to be further retained and acquiring a spatial–spectral consistency feature.

As mentioned above, there is complementary information among neighbor bands. To enhance the reliability of the intermediate results, we combine a consistency feature $\boldsymbol{F}_{t-1}^{\text{Nei}}$ from the previous band using a concatenation operation. Finally, we leverage the PixelShuffle operator to upscale the acquired feature to the target size and implement a global residual learning to reconstruct the intermediate SR band $\boldsymbol{I}_t^{\text{Nei}}$.

C. Context Network

In general, by taking advantage of the high similarity among the neighbor bands, recursive methods have achieved considerable performance with small memory budgets in single HIS. In addition, through a feedback mechanism, such methods introduce the context information to further improve the integrity of the information. As previously mentioned, both neighbor band information and context play critical roles in HIS. Nevertheless, the existing methods primarily focus on fully exploiting useful information from neighbor bands while neglecting effective integration of context information. Specifically, context information is often incorporated via a single-step operation, such as concatenation or addition. In this manner, all the components of the context information receive equal treatment, which introduces a significant amount of redundant features.

To this end, we purpose a context network to effectively use the complementary information within the context. As shown in Fig. 4, the context network is constructed by cascading multiple AE modules. Each AE module aims to learn the relationship between the current band $\boldsymbol{F}_t^{\text{Nei}}$ and its feedback context information $\boldsymbol{F}_{t-1}^{\text{Con}}$ at the feature level. It is introduced to encourage the network to explore more complementary information in the context and suppress the introduction of irrelevant information.

At the beginning, the current band feature $\boldsymbol{F}_t^{\text{Nei}}$ and its context information $\boldsymbol{F}_{t-1}^{\text{Con}}$ are fed into the context network and pass through the Res module to learn their original semantic features, i.e., $\boldsymbol{N}_j^t$ and $\boldsymbol{C}_j^t$, where $j = 1$. Then, to construct the correlation representation between $\boldsymbol{N}_1^t$ and $\boldsymbol{C}_1^t$, an AE

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:29:33 UTC from IEEE Xplore. Restrictions apply.

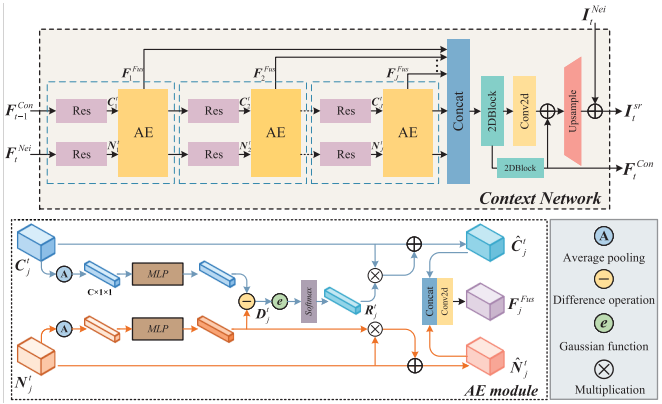


Fig. 4. Architecture of the context network (located above) and details of the AE module (located below). The contextual network is constructed by cascading multiple AE modules, with the input of the current spectral feature $\boldsymbol{F}_t^{\text{Nei}}$ and feedback contextual information $\boldsymbol{F}_{t-1}^{\text{Con}}$, aiming to iteratively enhance and combine the complementary information within the feedback feature $\boldsymbol{F}_{t-1}^{\text{Con}}$ through multiple AE modules. In the jth AE module, global semantic descriptors of the two inputs $\boldsymbol{N}_j^t, \boldsymbol{C}_j^t$ are obtained through the channel attention mechanism, and the initial channel-level difference relationship $\boldsymbol{D}_j^t$ is calculated by elementwise subtraction. Subsequently, the final calibrator $\boldsymbol{R}_j^t$ is obtained through Gaussian function mapping and a softmax operation for subsequent AE, resulting in an enhanced fused feature $\boldsymbol{F}_j^{\text{fus}}$ and two enhanced feature maps $\hat{\boldsymbol{N}}_j^t, \hat{\boldsymbol{C}}_j^t$.

module is applied, as shown at the bottom of Fig. 4. To set up a channelwise relationship between them, two channel attentions are first used in parallel. Specifically, an average pooling operator followed by a multilayer perceptron (MLP) is conducted to produce global descriptors of the two inputs. Subsequently, using an elementwise subtraction operation, we obtain a channel-level difference relationship between the current band and its context, which can be expressed as

$$\boldsymbol{D}_1^t = \mathcal{M}\big(\mathcal{A}\big(\boldsymbol{N}_1^t\big)\big) - \mathcal{M}\big(\mathcal{A}\big(\boldsymbol{C}_1^t\big)\big) \tag{8}$$

where $\boldsymbol{D}_1^t$ represents the correlation difference between the current band and its context information, $\mathcal{A}$ is the average pooling operation, and $\mathcal{M}$ denotes the MLP network.

As the difference might be negative or positive, we map it to a meaningful descriptor by a Gaussian function

$$\boldsymbol{R}_1^t = \mathcal{S}\big(\varphi\big(\boldsymbol{D}_1^t\big)\big) \tag{9}$$

where $\varphi(x) = \exp(-(x^2/\sigma^2))$, which maps the obtained discrepancy $\boldsymbol{D}_1^t$ to a nonnegative representation. The result is not sensitive to σ, and we use $\sigma = 3$ in our experiments. In addition, under a softmax operation ($\mathcal{S}$), the gap between the discrepancy values is further enlarged, resulting in the final recalibrator $\boldsymbol{R}_1^t$. For each value of $\boldsymbol{R}_1^t$, it should be enhanced if the previous context feature $\boldsymbol{F}_{t-1}^{\text{Con}}$ has a high correlation with the current band information $\boldsymbol{F}_t^{\text{Nei}}$, which prompts the network to learn more complementary information from the context. In practice, to acquire an enhanced context feature which contains richer complementary information, the input $\boldsymbol{C}_1^t$ is multiplied with the recalibrator $\boldsymbol{R}_1^t$, then a residual connection is used to preserve detail information. Formulaically, the enhanced context map $\hat{\boldsymbol{C}}_1^t$ could be expressed as

$$\hat{\boldsymbol{C}}_1^t = \boldsymbol{R}_1^t \times \boldsymbol{F}_{t-1}^{\text{Con}} + \boldsymbol{F}_{t-1}^{\text{Con}}. \tag{10}$$

Similarly, we also adopt a channel attention operation to obtain the enhanced map of the input current band feature $\boldsymbol{N}_1^t$. By this way, it assists the network focus on more useful information, i.e.,

$$\hat{\boldsymbol{N}}_1^t = \mathcal{M}\big(\mathcal{A}\big(\boldsymbol{N}_1^t\big)\big) \times \boldsymbol{N}_1^t + \boldsymbol{N}_1^t \tag{11}$$

where $\hat{\boldsymbol{N}}_1^t$ is the enhanced map of the current band.

Finally, the two enhanced maps are fused for subsequent analyses, i.e.,

$$\boldsymbol{F}_1^{\text{fus}} = 2\mathcal{D}\big(\mathcal{C}\big(\hat{\boldsymbol{N}}_1^t, \hat{\boldsymbol{C}}_1^t\big)\big) \tag{12}$$

where $\boldsymbol{F}_1^{\text{fus}}$ is the enhanced fused feature. $2\mathcal{D}$ refers to the 2-D convolution and $\mathcal{C}$ represents a concatenation operation.

In summary, the proposed AE module produces an enhanced fused feature $\boldsymbol{F}_j^{\text{fus}}$ and two enhanced feature maps $\hat{\boldsymbol{N}}_j^t, \hat{\boldsymbol{C}}_j^t$ corresponding to its two original inputs. To further enhance the content perception capability, several AE modules are connected in series by the Res module. Passing through the Res module, the previous enhanced maps $\hat{\boldsymbol{N}}_j^t, \hat{\boldsymbol{C}}_j^t$ are fed into the next AE module to generate more effective enhanced maps $\hat{\boldsymbol{N}}_{j+1}^t, \hat{\boldsymbol{C}}_{j+1}^t$, where $j = 1, 2, \ldots, J-1$. Here, J is the number of AE modules. Meanwhile, J enhanced fused features can be obtained, and we fuse them together for subsequent reconstruction, i.e.,

$$\boldsymbol{F}^{\text{fus}} = \mathcal{C}\big(\boldsymbol{F}_1^{\text{fus}}, \boldsymbol{F}_2^{\text{fus}}, \ldots, \boldsymbol{F}_J^{\text{fus}}, \hat{\boldsymbol{N}}_J^t, \hat{\boldsymbol{C}}_J^t\big) \tag{13}$$

where $\boldsymbol{F}^{\text{fus}}$ is the final enhanced fused feature. Next, similar to the neighbor network, we obtain a semantic feature $\boldsymbol{F}_t^{\text{Con}}$ as the context information of the next iteration via a series of 2-D convolution operations. Meanwhile, under the action of upsampling operation and residual learning, the final reconstruction result $\boldsymbol{I}_t^{\text{sr}}$ is produced.

The proposed AE module draws on the experience of SA-Gate [60]. However, different from the SA-Gate, the AE module is designed to enhance complementary information in the context based on the content of the current band, rather than performing cross-enhancement between the inputs. In addition, we explicitly construct the correlation representation between input features, which can effectively guide the network to learn more useful features. Moreover, the AE module can be applied in other serialized data processing areas to enhance adaptability to the context.

D. Loss Function

In the existing studies, L_1 and L_2 losses are fundamental and commonly used loss functions. The L_2 loss, involving squared calculations, typically encourages excessive smoothing of outliers. In contrast, the L_1 loss is more sensitive to errors and can maintain good convergence during the training phase. Therefore, an L_1 loss function is adopted to jointly optimize the neighbor and final SR results, which can be expressed as

$$\mathcal{L} = \left\|\boldsymbol{I}_t^{\text{sr}} - \boldsymbol{I}_t^{\text{hr}}\right\|_1 + \lambda\left\|\boldsymbol{I}_t^{\text{Nei}} - \boldsymbol{I}_t^{\text{hr}}\right\|_1 \tag{14}$$

where $\boldsymbol{I}_t^{\text{hr}}$ is the ground truth (GT) of the tth spectral band $\boldsymbol{I}_t^{\text{lr}}$. The parameter λ represents the tradeoff of the two subnetworks, regulating the weight of context information enhancement. A small value may cause the network to focus much on content exploration in the neighbor group. On the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:29:33 UTC from IEEE Xplore. Restrictions apply.

TABLE I
QUANTITATIVE PERFORMANCE OF DIFFERENT LOSS FUNCTIONS ON THE CAVE DATASET AT THE SCALE FACTOR 2

$\mathcal{L}_{Nei}$	$\mathcal{L}_{Fin}$	PSNR	SSIM	SAM
L_1	L_1	45.805	0.9742	2.187
L_1	L_2	45.781	0.9741	2.195
L_2	L_1	45.788	0.9742	2.190
L_2	L_2	45.783	0.9741	2.198
$\times$	L_1	45.697	0.9741	2.211

other hand, for the two-stage method proposed in this article, a larger λ can prevent the network from trapping in local optima. Accordingly, we set λ as 10 in the following experiments.

In Table I, we present the reconstruction performance when using different loss functions. $\mathcal{L}_{\text{Nei}}$ and $\mathcal{L}_{\text{Fin}}$ are optimization functions for the neighbor and final SR results, respectively. It is evident that the use of the L_1 loss function yields the best results, as it effectively penalizes small errors and is more suitable for our algorithm. This suggests that the effectiveness of intermediate results is beneficial for obtaining final satisfactory reconstruction outcomes.

IV. EXPERIMENTAL STUDIES

In this section, we conduct experiments from both qualitative and quantitative perspectives to verify the effectiveness of our method on five benchmark hyperspectarl image datasets, i.e., CAVE,[1] Harvard,[2] Foster,[3] Salinas,[4] and Pavia Centre[5] datasets. First, we briefly introduce the datasets' construction and experimental implementation details. Then, we represent a series of experiments to verify the effectiveness of different modules. Finally, we compare the proposed method with other state-of-the-art peers.

A. Implementation Details

As for the three nature hyperspectral image datasets CAVE, Harvard, and Foster, each dataset contains multiple independent hyperspectral images. For example, the CAVE dataset contains 32 hyperspectral images from different scenes, such as food, drinks, and paints. Each hyperspectral image has a spatial resolution of 512 $\times$ 512 with 31 bands. Besides, the Harvard and Foster datasets include 70 and 30 images, respectively, and the size of them is 1392 $\times$ 1040 $\times$ 31 and 1204 $\times$ 1344 $\times$ 33. In our experiments, we use 80% of the data for training and 20% for testing. The Salinas and Pavia Centre datasets are remote sensing ones, comprising 204 and 103 bands, respectively, with spatial sizes of 512 $\times$ 217 and 1096 $\times$ 715. The bottom part with 128 $\times$ 217 spatial size of the Salinas dataset is cropped to form testing data and the rest is used as a training set. Likewise, in the Pavia Centre dataset, the top 968 $\times$ 715 pixels are used as the training set, and the rest area is assigned for a test.

[1]http://www1.cs.columbia.edu/CAVE/databases/multispectral/

[2]http://vision.seas.harvard.edu/hyperspec/explore.html

[3]https://personalpages.manchester.ac.uk/staff/d.h.foster/Local_Illumination_HSIs/Local_Illumination_HSIs_2015.html

[4]https://www.sal.t.u-tokyo.ac.jp/hyperdata/

[5]https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

As for the training images, we apply some data augmentation methods to expand the training set, including spatial scaling (1, 0.75, 0.5), randomly rotating (90°, 180°, 270°), and horizontally flipping. We randomly select 20 patches from each hyperspectral image as the HR images. Then, using the bicubic interpolation method, these patches are downsampled to obtain the corresponding LR images. The spatial resolution of each LR image is 32 $\times$ 32 $\times$ L, where L is the number of bands. For the test set, we extract 512 $\times$ 512 pixels from the top left of each image. According to the scale factors, the corresponding LR images are generated by bicubic downsampling. The Salinas dataset is a remote sensing one, which only contains a hyperspectral image. We randomly cropped 70 patches in the training image. The same data augmentation methods are used here to expand the training set.

For the training phase, all the convolution layers have 64 filters for final comparison. The size of each filter is set to 3 $\times$ 3. The ADAM optimizer is adopted with the default values of β_1 and β_2. The initial learning rate is set to 0.00001, which is reduced by ten times every 20 epochs. We choose the batch size of 64, and the total number of epochs is 70. In addition to pursuing powerful performance, the computational efficiency of models has received increasing attention, leading to the recent rise of lightweight SR networks. Therefore, based on the proposed MESR, we construct its corresponding efficient version called MESR-S, where the number of filters in all the convolutional layers is set to 32. The proposed algorithms are conducted on the PyTorch platform with NVIDIA GeForce RTX 3090 GPU.

B. Comparison on Nature Datasets

In this section, we compare the proposed algorithm against eight previous methods, including a basic interpolation method, bicubic, two typical nature image SR methods, EDSR [34] and SwinIR [41], and five single HIS methods, GDRRN [52], 3DFCNN [30], MCNet [22], EUNet [53], TC-HISRNet [54], MDFL [25], ERCSR [24], and SFCSR [28]. The experiments are conducted on the three benchmark hyperspectral image datasets, which include three nature hyperspectral image datasets, i.e., CAVE, Harvard, and Foster. Three widely used picture quality indexes are used for qualitative evaluation, including mean peak signal-to-noise ratio (PSNR), spectral angle mapper (SAM), and structure similarity index measure (SSIM). We demonstrate the performance of our method by quantitative evaluation and qualitative comparison.

1) Quantitative Evaluation: Table II represents quantitative evaluation results of the proposed MESR at different scale factors. The best results are marked in red and the second best ones are marked in blue. As shown in the table, our method is superior to other competitors at all scales on the CAVE dataset. Specifically, the bicubic method achieves the worst performance, especially for PSNR and SSIM. As for EDSR, SwinIR, and GDRRN, they focus on exploring spatial feature with 2-D convolutions. Although they make advances in spatial reconstruction accuracy, referring to PSNR and SSIM, they perform poorly in spectral fidelity (SAM). Due to the lack of spatial feature extraction ability, 3DFCNN obtains lower

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:29:33 UTC from IEEE Xplore. Restrictions apply.

TABLE II

QUANTITATIVE EVALUATION WITH OTHER STATE-OF-THE-ART COMPETITORS AT DIFFERENT SCALES ON THE CAVE, HARVARD, AND FOSTER DATASETS BY PARAMETER NUMBER, MEMORY USAGE, AND AVERAGE PSNR/SSIM/SAM

Method	Scale	Param.	Memory usage	FLOPs	Time	CAVE			Harvard			Foster		
						PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM
Bicubic [2D]	×2	-	-	-	-	40.762	0.9622	2.647	43.650	0.9739	2.358	55.125	0.9976	4.393
EDSR [34]	×2	1402.143k	1417M	191GFLOPs	0.609s	44.185	0.9728	2.522	45.434	0.9765	2.429	57.008	0.9983	4.742
SwinIR [41]	×2	985.864k	2215M	124GFLOPs	1.713s	42.088	0.9678	3.139	43.968	0.9676	2.837	52.848	0.9958	6.929
GDRRN [52]	×2	218.880k	1731M	183GFLOPs	0.699s	41.943	0.9653	3.808	45.070	0.9757	2.459	56.674	0.9983	4.392
3DFCNN [30]	×2	39.405k	4319M	120GFLOPs	0.625s	42.899	0.9674	2.447	44.554	0.9753	2.335	58.166	0.9982	6.534
MCNet [22]	×2	1928.334k	10177M	7850GFLOPs	1.152s	45.332	0.9739	2.211	46.361	0.9819	2.198	59.054	0.9988	4.062
EUNet [53]	×2	550.586k	1902M	349GFLOPs	0.573s	43.550	0.9725	2.398	45.811	0.9796	2.305	57.496	0.9985	4.243
TC-HISRNet [54]	×2	8230.364k	7406M	2030GFLOPs	5.559s	45.569	0.9736	2.227	46.515	0.9825	2.182	59.056	0.9988	4.049
MDFL [25]	×2	1679.524k	18734M	7260GFLOPs	0.811s	45.245	0.9738	2.224	46.304	0.9822	2.190	58.858	0.9987	4.054
ERCSR [24]	×2	1348.742k	10699M	5920GFLOPs	0.955s	45.320	0.9741	2.248	46.393	0.9819	2.199	58.595	0.9987	4.064
SFCSR [28]	×2	1848.882k	2577M	9052GFLOPs	1.568s	45.324	0.9739	2.211	46.440	0.9822	2.195	58.867	0.9987	4.176
MESR [Ours]	×2	3865.850k	2423M	16461GFLOPs	1.633s	45.805	0.9742	2.187	46.542	0.9826	2.173	59.071	0.9988	4.049
MESR-S [Ours]	×2	973.466k	1708M	4154GFLOPs	1.359s	45.553	0.9742	2.204	46.487	0.9825	2.173	59.056	0.9987	4.092
Bicubic [2D]	×3	-	-	-	-	37.535	0.9322	3.504	40.681	0.9535	2.731	50.942	0.9936	5.350
EDSR [34]	×3	1586.783k	1909M	100GFLOPs	0.429s	40.725	0.9514	3.090	42.226	0.9563	2.916	52.403	0.9954	6.122
SwinIR [41]	×3	1069.719k	2037M	64GFLOPs	1.224s	38.921	0.9423	3.795	41.526	0.9535	3.077	50.001	0.9925	7.996
GDRRN [52]	×3	218.880k	2251M	81GFLOPs	0.587s	39.853	0.9473	3.142	42.022	0.9604	2.743	52.760	0.9957	5.312
3DFCNN [30]	×3	39.405k	4815M	52GFLOPs	0.679s	39.328	0.9421	2.891	41.257	0.9540	2.702	55.167	0.9962	6.935
MCNet [22]	×3	2038.926k	6569M	3670GFLOPs	1.166s	41.236	0.9526	2.799	43.385	0.9662	2.627	54.191	0.9970	5.087
EUNet [53]	×3	632.506k	1832M	215GFLOPs	0.340s	40.362	0.9502	3.041	42.939	0.9638	2.685	53.243	0.9961	5.213
TC-HISRNet [54]	×3	8415.324k	3396M	782GFLOPs	2.070s	41.422	0.9506	2.812	43.532	0.9674	2.626	54.868	0.9969	5.119
MDFL [25]	×3	1790.116k	14744M	3730GFLOPs	0.535s	41.371	0.9527	2.763	43.366	0.9669	2.613	54.594	0.9967	5.103
ERCSR [24]	×3	1459.334k	8793M	3140GFLOPs	0.787s	41.171	0.9517	2.784	43.417	0.9657	2.634	54.627	0.9967	5.052
SFCSR [28]	×3	1269.842k	2177M	4340GFLOPs	1.238s	41.309	0.9527	2.814	43.357	0.9663	2.612	54.856	0.9968	5.119
MESR [Ours]	×3	4235.770k	2129M	7905GFLOPs	1.679s	41.620	0.9526	2.746	43.542	0.9682	2.596	54.876	0.9969	5.045
MESR-S [Ours]	×3	1066.266k	1894M	1984GFLOPs	1.109s	41.463	0.9526	2.762	43.436	0.9670	2.612	54.223	0.9966	5.149
Bicubic [2D]	×4	-	-	-	-	35.749	0.9073	3.927	38.776	0.9356	2.949	48.265	0.9876	5.994
EDSR [34]	×4	1549.855k	1895M	74GFLOPs	0.548s	38.647	0.9290	3.603	40.280	0.9434	3.071	49.684	0.9916	6.792
SwinIR [41]	×4	1187.116k	1963M	38GFLOPs	1.269s	37.311	0.9204	4.301	39.736	0.9392	3.262	48.293	0.9888	8.222
GDRRN [52]	×4	218.880k	2249M	46GFLOPs	0.446s	37.478	0.9209	4.489	40.107	0.9416	3.062	49.895	0.9914	5.989
3DFCNN [30]	×4	39.405k	4815M	28GFLOPs	0.496s	37.251	0.9177	3.435	39.290	0.9358	2.884	51.818	0.9915	7.554
MCNet [22]	×4	2174.094k	5339M	2230GFLOPs	1.065s	39.069	0.9320	3.235	41.334	0.9511	2.875	52.303	0.9943	5.721
EUNet [53]	×4	616.250k	1942M	179GFLOPs	0.385s	38.521	0.9092	3.467	40.898	0.9481	2.923	50.483	0.9925	5.754
TC-HISRNet [54]	×4	8378.332k	2136M	449GFLOPs	1.466s	39.523	0.9323	3.202	41.400	0.9520	2.861	51.782	0.9937	5.949
MDFL [25]	×4	1925.284k	12404M	2520GFLOPs	0.414s	39.220	0.9328	3.204	41.280	0.9515	2.871	52.037	0.9940	5.735
ERCSR [24]	×4	1594.502k	7473M	2180GFLOPs	0.859s	39.222	0.9323	3.180	41.333	0.9517	2.859	51.478	0.9933	5.631
SFCSR [28]	×4	1232.850k	2063M	2852GFLOPs	0.725s	39.235	0.9322	3.202	41.169	0.9512	2.862	51.822	0.9940	6.324
MESR [Ours]	×4	4161.786k	2059M	5301GFLOPs	2.026s	39.592	0.9324	3.176	41.430	0.9521	2.861	51.725	0.9934	5.604
MESR-S [Ours]	×4	1047.706k	1644M	1333GFLOPs	1.128s	39.384	0.9314	3.205	41.402	0.9518	2.865	51.750	0.9935	5.635
Bicubic [2D]	×8	-	-	-	-	31.807	0.8485	5.291	35.122	0.8980	3.326	43.109	0.9595	7.854
EDSR [34]	×8	1697.567k	1377M	48GFLOPs	0.578s	31.551	0.8233	10.667	31.522	0.8922	3.326	44.000	0.9697	8.483
SwinIR [41]	×8	985.864k	1905M	16GFLOPs	0.880s	33.548	0.8653	5.768	36.479	0.9090	3.519	43.245	0.9644	10.761
GDRRN [52]	×8	218.880k	1731M	11GFLOPs	0.451s	33.005	0.8535	7.443	36.429	0.9088	3.369	44.234	0.9683	7.462
3DFCNN [30]	×8	39.405k	4319M	7GFLOPs	0.669s	33.097	0.8602	4.220	35.710	0.8900	3.168	45.594	0.9667	8.579
MCNet [22]	×8	2960.526k	4139M	779GFLOPs	0.824s	35.500	0.8850	4.284	36.804	0.9144	3.278	45.702	0.9759	7.591
EUNet [53]	×8	681.914k	1782M	136GFLOPs	0.389s	35.022	0.8800	4.504	37.043	0.8903	3.298	44.790	0.9726	7.211
TC-HISRNet [54]	×8	8526.300k	2028M	127GFLOPs	0.733s	35.592	0.8830	4.328	37.180	0.9161	3.316	45.222	0.9732	7.644
MDFL [25]	×8	2711.716k	10206M	1280GFLOPs	0.329s	35.493	0.8848	4.345	37.071	0.9156	3.323	45.765	0.9763	7.385
ERCSR [24]	×8	2380.934k	5806M	1200GFLOPs	0.899s	35.494	0.8846	4.305	37.170	0.9163	3.265	45.365	0.9740	7.298
SFCSR [28]	×8	1380.818k	1943M	1333GFLOPs	0.636s	35.413	0.8840	4.319	37.173	0.9163	3.266	45.225	0.9732	7.714
MESR [Ours]	×8	4457.722k	2021M	2542GFLOPs	1.263s	35.672	0.8868	4.333	37.230	0.9164	3.309	45.232	0.9736	7.162
MESR-S [Ours]	×8	1121.946k	1634M	620GFLOPs	1.155s	35.525	0.8834	4.391	37.188	0.9164	3.311	45.145	0.9735	7.203

PSNR and SSIM. Although having the smallest number of parameters, it requires a larger memory budget compared with these 2-D convolution methods.

The spatial–spectral structure can be well-preserved after introducing 3-D convolutions. As a result, on the three nature datasets, 3-D-based networks (MDFL, MCNet, ERCSR, SFCSR, and MESR) achieve better performance than 2-D-based methods in both spatial reconstruction accuracy and spectral fidelity. However, the use of 3-D convolutions often causes a dramatic increase in memory budgets. Besides, MCNet, MDFL, and ERCSR take all the spectral bands as input for processing simultaneously, further increasing the memory space requirement for devices in both the training and testing processes. By leveraging the long-range dependency exploration capability of the transformer, TC-HISRNet achieves desirable results. However, the limited number of samples in the hyperspectral dataset constrains its performance. By taking advantage of the high correlation among adjacent bands, SFCSR applies a recursive structure to super-resolve a hyperspectral image band-by-band. It attains considerable performance due to a further constraint on solution space. However, like most recursive networks [27], [29], it still ignores how to combine the feedback information effectively. As for MESR, by constructing the relationship between the current band with its feedback information, the information in the context with high complementarity is further enhanced. Consequently, overall, it performs best across all the metrics, especially in PSNR and SSIM. Moreover, compared with other 3D-based competitors, it requires the lowest memory budget. As for the Foster dataset, although the optimality is not achieved at large scale factors, MESR also achieves comparable performance and has an evident advantage in spectral structure preservation. Through the above description, it can be proved that our method performs favorably against

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:29:33 UTC from IEEE Xplore. Restrictions apply.

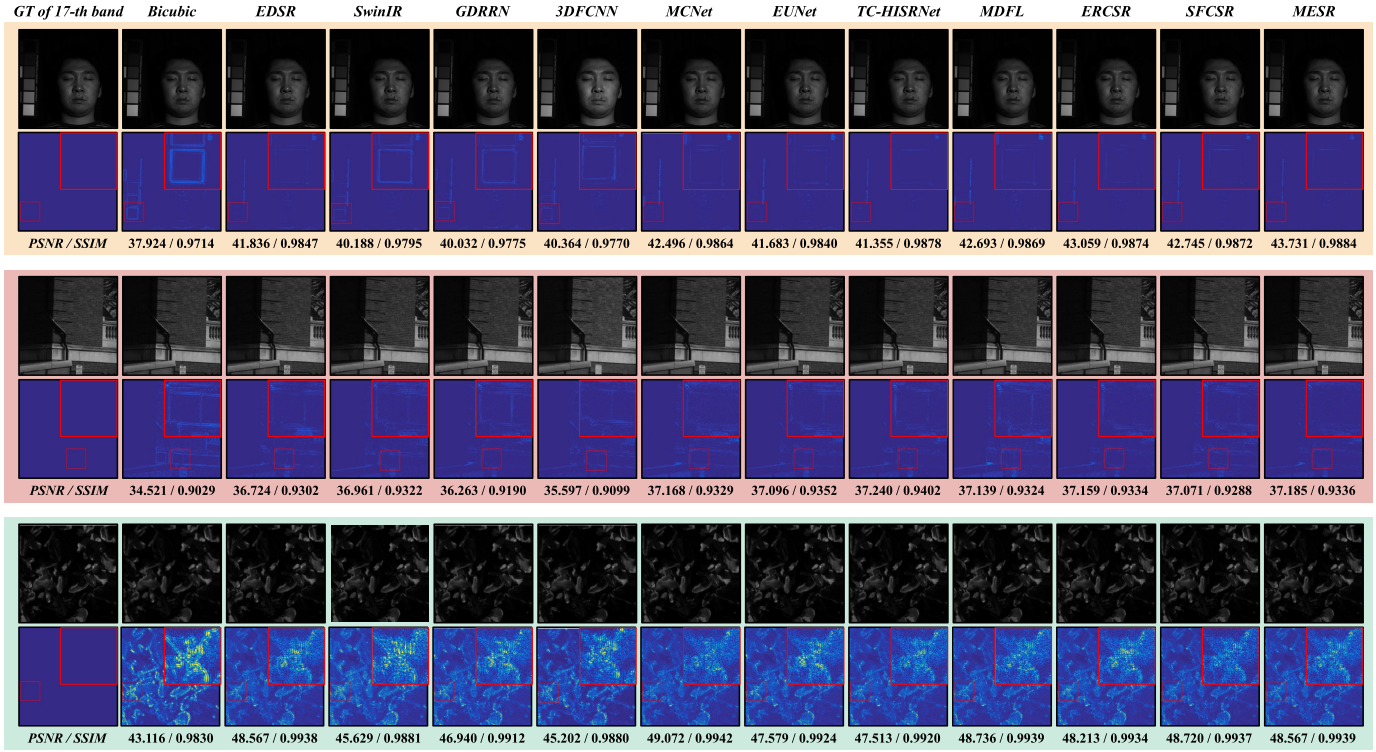


Fig. 5. Reconstructed results and corresponding absolute error maps generated by different methods. From top to bottom are the visual results of the image face_ms in the CAVE dataset, imgb2 in the Harvard dataset, and Bom_Jesus_Bush in the Foster dataset. The bluer the area in the absolute error map, the smaller the absolute error with GT.

the other state-of-the-art peers in terms of reconstruction quality and memory efficiency.

Furthermore, due to the multiple recursive operations involved, recursive algorithms such as SFCSR and MESR attain high FLOPs. Therefore, we further develop an efficient version of the proposed method, called MESR-S. Compared with other high-performing competitors, MESR-S not only demonstrates comparable performance but also exhibits fewer parameters, higher memory efficiency, and lower computational complexity (FLOPs).

2) Qualitative Comparison: To qualitatively evaluate the performance of the proposed method, we present the visual results of the 17th band in one image from the testing set of each dataset for a scale factor $\times 4$, as shown in the first row of each dataset in Fig. 5. To observe the quality of the reconstruction results more intuitively, we also display the absolute error map between the reconstruction result and GT image. The bluer the error map, the better the reconstruction performance. The PSNR and SSIM of the corresponding hyperspectral image for each method are reported at the bottom. As shown in the maps, our method has a lower absolute error overall (closer to blue). In addition, the edges in some areas are shallower or disappear for the absolute error map of the MESR. These visual results further demonstrate the excellent ability of our method in HIS.

To visually compare spectral reconstruction performance, the first column in Fig. 6 depicts the mean absolute error (MAE) plots between the reconstructed and GT bands. Compared with showing the spectral curves of some positions, the MAE plot is more representative. From these plots, it can be observed that the curves of our method display minimal values on three datasets. This proves that MESR is superior

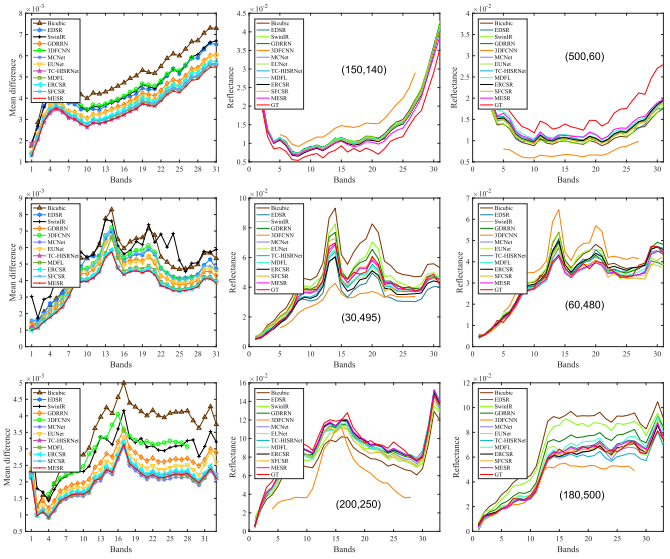


Fig. 6. Mean absolute error curves (column one) and spectral reconstruction results corresponding to two pixel positions (columns two and three) on the test images of the three datasets. The test image hairs in CAVE, imga3 in Harvard, and Bom_Jesus_Bush in Foster are performed from top to bottom.

to other comparing methods in spectralwise consistent restoration. In addition, we randomly select some pixels of the test image on each dataset for visual comparison. As shown in the second and third columns in Fig. 6, the reconstructed spectra of the proposed method are closer to GT's, which indicates that our MESR obtains lower spectral distortion. Through the above qualitative comparison, it can be further indicated that the proposed method not only has remarkable performance in spatial information reconstruction but also performs well in spectral consistent preservation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:29:33 UTC from IEEE Xplore. Restrictions apply.

TABLE III

QUANTITATIVE EVALUATION ON THE SALINAS DATASET AT THE SCALE FACTORS ×2 AND ×4

Method	×2				×4			
	Memory	PSNR	SSIM	SAM	Memory	PSNR	SSIM	SAM
Bicubic	-	42.050	0.9862	0.700	-	38.105	0.9673	0.902
GDRRN	1781M	42.048	0.9862	0.701	1795M	38.104	0.9673	0.903
3DFCNN	3113M	38.369	0.9692	1.840	3113M	37.992	0.9668	0.973
MCNet	5246M	43.993	0.9915	0.647	3558M	39.480	0.9785	0.840
EUNet	2830M	42.376	0.9871	0.699	2628M	38.248	0.9682	0.900
TC-HISRNet	1960M	43.921	0.9912	0.713	1846M	39.226	0.9778	0.930
MDFL	8260M	44.013	0.9915	0.642	5326M	39.463	0.9785	0.840
ERCSR	5805M	43.893	0.9915	0.639	4347M	39.400	0.9782	0.867
SFCSR	1793M	43.124	0.9891	0.765	1759M	39.333	0.9768	0.918
MESR	1849M	44.015	0.9915	0.676	1837M	39.522	0.9782	0.860

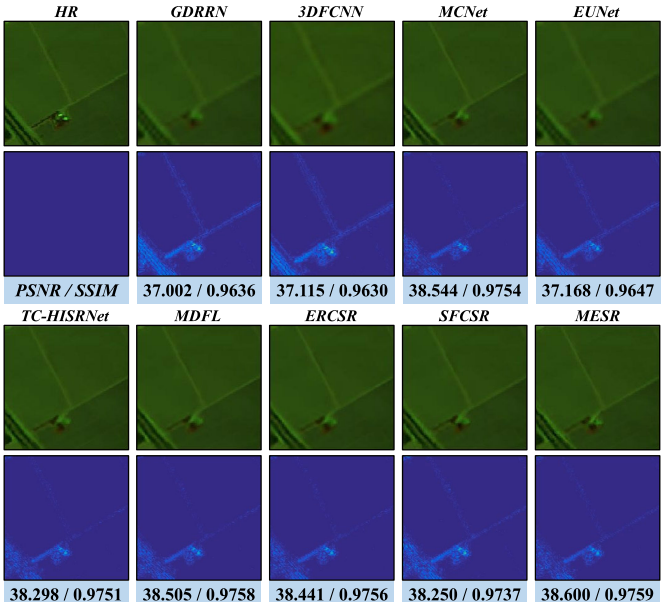


Fig. 7. Visualization results of a part of the test image in the Salinas dataset with spectral bands 90-20-180 as R-G-B and absolute error maps at the scale factor ×4 (the first row). The corresponding absolute error maps are shown in the second row.

C. Comparison on the Salinas Dataset

To evaluate the performance of the proposed MESR on the remote sensing dataset, we make a comparison with the state-of-the-art methods on the Salinas dataset. Since a limited spatial size of this dataset, a larger upscale factor often results in the model unfitting, which makes the results unstable. We perform quantitative experiments with scale factors ×2 and ×4 in this section, as shown in Table III. In comparison, MCNet, TC-HISRNet, MDFL, ERCSR, and MESR exhibit superior performance, with MCNet, MDFL, and MESR displaying more pronounced advantages. However, it is evident that the memory efficiency of MCNet, MDFL, and ERCSR is significantly poor. In addition, due to the recursive architecture, SFCSR and MESR can obtain good performance while requiring less memory budget. To sum up, the proposed method has obvious advantages in both reconstruction performance and memory occupation.

In Fig. 7, we visualize the reconstruction result of one test image at scale factor ×4. The absolute error maps corresponding to each method are displayed below. It is obvious that bicubic, EDSR, SwinIR, GDRRN, 3DFCNN, and EUNet produce more blurry edges on their reconstructed images, and the proposed method has shallower edges. This demonstrates the superiority of MESR in spatial information reconstruction. Mean absolute error curves at scale factor ×4 are performed in Fig. 8. The corresponding curve for MESR is obviously at the bottom. This means that the proposed method has excellent capability in spatial–spectral consistency maintenance. In addition, the spectral curves at a randomly selected pixel are also drawn in Fig. 8. MESR presents more accurate results on spectral reconstruction that is closer to GT's.

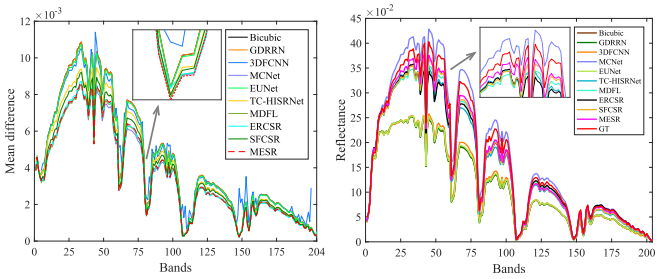


Fig. 8. (Left) Mean absolute error curves of one test hyperspectral image and (right) spectral curves at a randomly selected position in the Salinas dataset at the scale factor ×4 (the first row). The corresponding absolute error maps are shown in the second row.

TABLE IV

QUANTITATIVE EVALUATION OF STATE-OF-THE-ART ALGORITHMS THAT DESIGNED FOR HIS BY AVERAGE PSNR/SSIM/SAM FOR DIFFERENT SCALE FACTORS ON THE PAVIA CENTRE DATASET

Scale	Index	Bicubic	GDRRN	3DFCNN	MCNet	EUNet	TC-HISRNet	MDFL	ERCSR	SFCSR	MESR
×2	PSNR	32.471	33.614	33.946	35.330	34.508	35.141	35.267	35.389	35.058	35.331
	SSIM	0.9026	0.9264	0.9348	0.9492	0.9385	0.9479	0.9486	0.9485	0.9466	0.9489
	SAM	4.109	4.134	3.669	3.561	3.796	3.640	3.564	3.483	3.661	3.603
×4	PSNR	27.790	28.379	28.604	28.875	28.838	28.615	28.889	28.920	28.765	28.923
	SSIM	0.7104	0.7503	0.7625	0.7777	0.7732	0.7675	0.7772	0.7765	0.7699	0.7767
	SAM	5.858	6.037	5.078	5.864	5.425	6.170	5.726	5.585	5.770	5.723
×8	PSNR	24.928	25.389	25.226	25.402	25.343	25.100	25.408	25.391	25.312	25.410
	SSIM	0.5306	0.5777	0.5573	0.5671	0.5657	0.5501	0.5670	0.5658	0.5597	0.5674
	SAM	7.619	7.129	6.938	7.724	7.519	8.716	7.734	7.782	7.784	7.690

D. Comparison on the Pavia Dataset

We further carry out experiments on Pavia Centre dataset to verify the performance of MESR on the remote sensing dataset, which are presented in Table IV. Underline represents the third Best. Taking a comprehensive analysis, the proposed MESR attains considerable reconstruction performance in both spatial and spectral information, particularly the spatial restoration at larger scales. Fig. 9 displays a visual representation of the reconstructed HSI data. The absolute error maps are also provided for a clear comparison of the recovery performance. It can be observed that the proposed method exhibits low error levels (closer to blue). Furthermore, Fig. 10 depicts the absolute error curves of the reconstructed results (left), as well as spectral curves at a randomly selected position (right). It is evident that MDFL, ERCSR, and MESR exhibit lower mean differences across all the spectral bands, and their spectral structure is closer to the GT's. The above qualitative and quantitative experimental results demonstrate the overall advantages of the proposed method in spatial reconstruction and spectral preservation.

We further conduct hyperspectral image classification experiments on the reconstructed results to further assess the performance of the proposed method. Specifically, the orig-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:29:33 UTC from IEEE Xplore. Restrictions apply.

TABLE V

CLASSIFICATION RESULTS OF THE SVM AND KNN FOR THE RECONSTRUCTED IMAGES OF DIFFERENT METHODS ON THE PAVIA CENTRE DATASET

No.	Categories	Sample	KNN										SVM									
			GT	GDRRN	3DFCNN	MCNet	EUNet	HISRNet	MDFL	ERCSR	SFCSR	MESR	GT	GDRRN	3DFCNN	MCNet	EUNet	HISRNet	MDFL	ERCSR	SFCSR	MESR
C1	Water	64470	0.9996	0.9981	0.9237	0.9988	1	0.9996	0.9994	0.9993	0.9994	0.9973	0.9999	1	0.9892	1	1	0.9995	0.9999	0.9998	0.9998	0.9990
C2	Tree	6816	0.9328	0.7965	0.4775	0.8515	0.8686	0.9251	0.8693	0.8702	0.9105	0.8989	0.9811	0.8906	0.4604	0.8573	0.9269	0.9329	0.9079	0.9208	0.9376	0.9540
C3	Meadow	2535	0.7210	0.5233	0.0429	0.5527	0.6275	0.6957	0.5926	0.6139	0.6663	0.6864	0.8100	0.9610	0.0823	0.9263	0.9192	0.8837	0.9175	0.9151	0.9014	0.8627
C4	Brick	2685	0.6906	0.4895	0.1703	0.5709	0.6460	0.6659	0.6195	0.6579	0.6769	0.7358	0.8164	0.7894	0.2410	0.7929	0.8009	0.7408	0.8388	0.8746	0.8905	0.9035
C5	Soil	6560	0.9499	0.8808	0.3716	0.9202	0.9041	0.9497	0.9336	0.9295	0.9555	0.9541	0.9698	0.9838	0.4144	0.9824	0.9530	0.9442	0.9779	0.9776	0.9762	0.9676
C6	Asphalt	8506	0.9400	0.4866	0.1618	0.5021	0.6247	0.6376	0.5695	0.5677	0.6086	0.6551	0.9770	0.5956	0.2657	0.6655	0.8321	0.7640	0.7112	0.7083	0.7669	0.7758
C7	Bitumen	7287	0.8985	0.7576	0.2368	0.7544	0.8087	0.8566	0.8380	0.8175	0.8493	0.6551	0.9375	0.9272	0.2912	0.9130	0.9127	0.9398	0.9268	0.9247	0.9433	0.9392
C8	Tile	42826	0.9920	0.9880	0.6621	0.9756	0.9841	0.9610	0.9732	0.9669	0.9726	0.9674	0.9955	0.9935	0.7990	0.9869	0.9786	0.9428	0.9833	0.9793	0.9814	0.9800
C9	Shallow	2782	0.9989	0.9244	0.0205	0.8507	0.8767	0.8979	0.8944	0.8778	0.8269	0.8401	0.9985	0.9244	0.0125	0.8452	0.8778	0.8951	0.894	0.8749	0.8291	0.8448
	AA		0.9026	0.7606	0.3408	0.7752	0.8156	0.8432	0.8099	0.8112	0.8296	0.8440	0.9429	0.8962	0.3951	0.8855	0.9112	0.8936	0.9064	0.9083	0.9140	0.9141
	OA		0.9727	0.9188	0.6720	0.9212	0.9375	0.9404	0.9328	0.9304	0.9396	0.9415	0.9851	0.9586	0.7542	0.9564	0.9649	0.9513	0.9625	0.9618	0.9667	0.9665
	Kappa		0.9611	0.8844	0.5469	0.8878	0.9109	0.9150	0.9043	0.9009	0.9139	0.9167	0.9787	0.9411	0.6474	0.9379	0.9500	0.9308	0.9466	0.9456	0.9526	0.9523

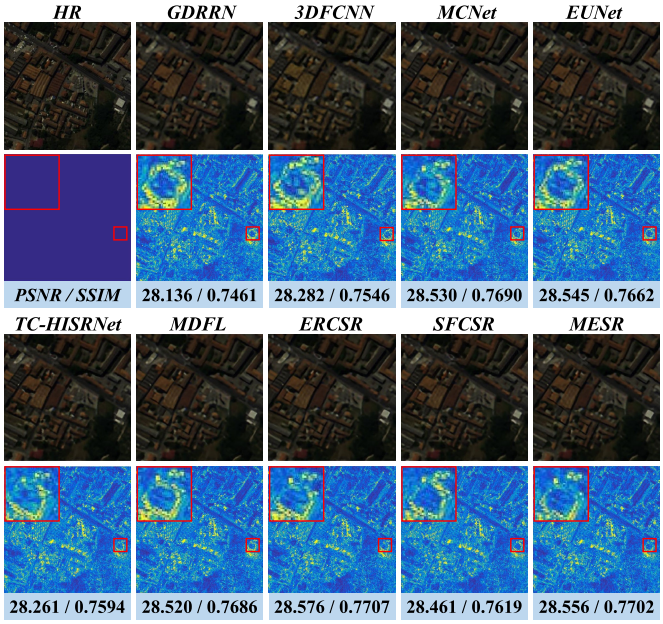


Fig. 9. Visualization results of a part of the test image in the Pavia Centre dataset with spectral bands 50-35-10 as R-G-B at the scale factor ×4.

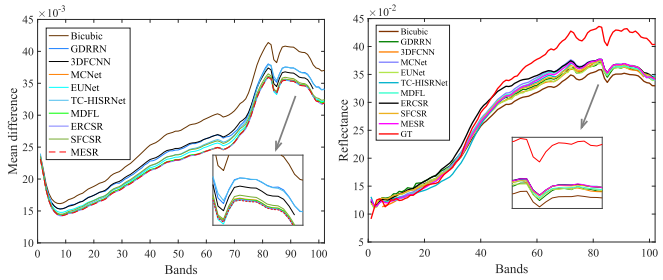


Fig. 10. (Left) Mean absolute error curves of one test hyperspectral image and (right) spectral curves at a randomly selected position in the Pavia Centre dataset at the scale factor ×4.

inal hyperspectral image with a size of 1096 × 715 × 102 is clipped to 1024 × 704 × 102 and then downsampled to 128 × 88 × 102. Subsequently, the downsampled image is super-resolved by the trained models at scale factor ×8, generating the reconstruction images with the size of 1024 × 704 × 102. The hyperspectral image classification experiments are conducted using the K-nearest neighbor (KNN) and support vector machine (SVM) algorithms. The training set is generated from the original hyperspectral data. In this study, we randomly select 2% samples for training and evaluate the classification performance on super-resolved images. Average accuracy (AA), overall accuracy (OA), and Kappa coefficient indices are used to evaluate classification performance. The quantitative classification results are presented in Table V (TC-HISRNet is denoted as HISRNet for convenience), while the classification map is shown in Fig. 11. In terms of classification performance, the results obtained from EUNet, TC-HISRNet, SFCSR, and MESR exhibit a remarkable superiority over other algorithms across all the evaluation indices. Notably, the proposed MESR displays a classification performance that is closer to the GT. Our findings provide further evidence that the hyperspectral images reconstructed by our proposed method attain the highest degree of spatial–spectral fidelity.

TABLE VI

STUDY OF THE NUMBER OF NEIFE AND 2DBLOCK MODULES ON THE CAVE DATASET AT THE SCALE FACTOR 2

Evaluation indices	2	3	4	5	6
PSNR	45.297	45.542	45.699	45.633	45.656
SSIM	0.9740	0.9739	0.9741	0.9739	0.9739
SAM	2.213	2.210	2.211	2.210	2.209

E. Model Analysis

1) Study of the Modules Number: For the proposed network, the number of NeiFE, 2DBlock, and AE modules could affect its performance. To determine how many modules are appropriate for the network, we analyze the changes on performance under different numbers of NeiFE, 2DBlock modules, and AE modules. Specifically, as shown in Table VI, on the CAVE dataset at the scale factor 2, we first set different numbers of the NeiFE and 2DBlock modules (M = 2, 3, 4, 5, 6) when only neighbor network exists to acquire a superior neighbor network structure, with the context network being discarded. It can be clearly seen that PSNR values have obvious improvement from 3 to 4, while SSIM and SAM values have little change. Thus, we set M as 4.

After that, we fix the structure of the neighbor network and vary the number of AE modules (J is set from 1 to 6) in the context network. The results are shown in Table VII. It can be seen that using AE modules improves the overall performance, particularly for PSNR, while the SSIM and SAM indexes remain almost unchanged when J is set to a small value (J = 1, 2). This is attributed to the limited modeling capacity of a small number of AE modules. As the number of AE modules increases, the overall results improve significantly (from 2 to 4) and then decrease slightly (from 4 to 6) and reach the best result at J = 4. It indicates that more AE mod-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:29:33 UTC from IEEE Xplore. Restrictions apply.

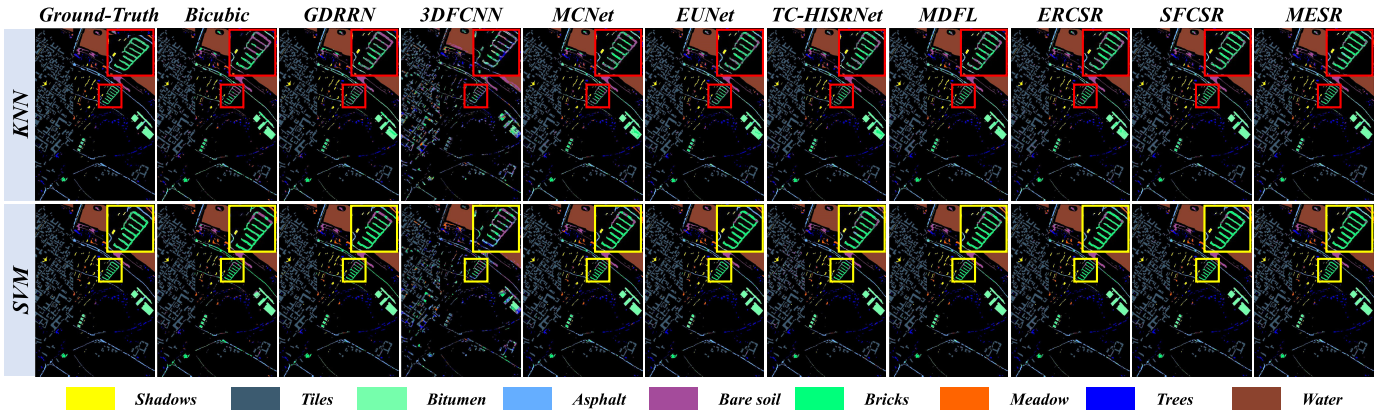


Fig. 11. Classification maps obtained by SVM and KNN methods on the reconstruction results at scale factor 8 on the Pavia Centre dataset.

TABLE VII

STUDY OF THE NUMBER OF AE MODULES ON THE CAVE DATASET AT THE SCALE FACTOR 2

evaluation indices	1	2	3	4	5	6
PSNR	45.702	45.789	45.764	45.805	45.774	45.736
SSIM	0.9739	0.9739	0.9738	0.9742	0.9739	0.9741
SAM	2.213	2.210	2.196	2.187	2.198	2.199

TABLE VIII

STUDY OF THE DIFFERENT INPUT FORMS ON THE CAVE DATASET AT THE SCALE FACTOR 2

Input form	PSNR↑	SSIM↑	SAM↓
B1	45.505	0.9738	2.204
B3	45.650	0.9741	2.189
B5 (ours)	45.805	0.9742	2.187
B7	45.789	0.9742	2.186
Our - w/ OL_1	45.704	0.9740	2.192
Our - w/ OL_2	45.726	0.9740	2.205
Our - w/o grouping	45.693	0.9740	2.202
Our - w/o $order_1$	45.731	0.9740	2.196
Our - w/o $order_2$	45.671	0.9739	2.204

ules not only help enhance complementary information from the context but also inhibit the introduction of redundancy information to reduce interference with the spatial–spectral consistency information from the neighbor network. However, a large number of AE modules weaken the influence of neighbor information on the reconstruction results, resulting in poor performance ($J = 5, 6$). Therefore, for subsequent experiments, we set the number of NeiFE and 2DBlock modules to four in the neighbor network and deploy four AE modules in the context network, i.e., $M = 4$ and $J = 4$.

2) Influence of Input Forms: To fully explore the spatial and spectral information among the neighbor bands, in our work, five neighbor bands are taken as input and divided into three ordered subgroups for parallel processing (B5), in which the spectral structure is effectively preserved. In this section, we study the effect of different input forms on network performance. Specifically, we investigate the impact of different input forms on the CAVE datasets at scale factor 2, as shown in Table VIII. We first adopt 1, 3, and 7 bands as input, which are denoted as B1, B3, and B7, respectively. When only one band is used as input (B1), the band is duplicated twice to form three subgroups. As illustrated in Table VIII, using more neighbor bands as input yields better results for our method. Moreover, when the number of input bands is greater than 5, the PSNR value drops slightly due to the introduction of redundant information, limiting reconstruction accuracy in the spatial domain. Besides, more spectral bands inevitably require more memory budgets. Therefore, we empirically set the number of input bands to 5.

To further verify the effectiveness of the proposed grouping strategy, as expressed in (4), we compare it with different variants of the proposed grouping method. We first append the current band to other two subgroups, i.e., $[\boldsymbol{I}^{\text{lr}}_{t-2}, \boldsymbol{I}^{\text{lr}}_{t-1}, \boldsymbol{I}^{\text{lr}}_{t}]$, $[\boldsymbol{I}^{\text{lr}}_{t}]$, $[\boldsymbol{I}^{\text{lr}}_{t}\ \boldsymbol{I}^{\text{lr}}_{t+1}, \boldsymbol{I}^{\text{lr}}_{t+2}]$, where there is overlap between them, as shown in Table VIII, denoted as "Our - w/OL_1." Besides, we construct another input form with overlapping bands, i.e., $[\boldsymbol{I}^{\text{lr}}_{t-2}, \boldsymbol{I}^{\text{lr}}_{t-1}]$, $[\boldsymbol{I}^{\text{lr}}_{t-1}, \boldsymbol{I}^{\text{lr}}_{t}, \boldsymbol{I}^{\text{lr}}_{t+1}]$, $[\boldsymbol{I}^{\text{lr}}_{t+1}, \boldsymbol{I}^{\text{lr}}_{t+2}]$, which is donated as "Our - w/OL_2." Their performance declines significantly, especially for the SAM of "Our - w/OL_2" (+0.018). This is because when the current band merges with the neighbor band on one side, the neighbor band information of another side is also introduced, causing a loss of spectral structure preservation in parallel processing. Then, we duplicate the neighbor bands and feed them into three branches without using any grouping strategy, denoted as "Our - w/o grouping." The performance shows a significant decrease. In addition, we disarrange the order of the input bands, i.e., $[\boldsymbol{I}^{\text{lr}}_{t}]$, $[\boldsymbol{I}^{\text{lr}}_{t-2}, \boldsymbol{I}^{\text{lr}}_{t-1}]$, $[\boldsymbol{I}^{\text{lr}}_{t+1}, \boldsymbol{I}^{\text{lr}}_{t+2}]$, and $[\boldsymbol{I}^{\text{lr}}_{t-2}, \boldsymbol{I}^{\text{lr}}_{t}, \boldsymbol{I}^{\text{lr}}_{t+2}]$, $[\boldsymbol{I}^{\text{lr}}_{t}]$, $[\boldsymbol{I}^{\text{lr}}_{t-1}, \boldsymbol{I}^{\text{lr}}_{t}, \boldsymbol{I}^{\text{lr}}_{t+1}]$, which are donated as "Our - w/o $order_1$" and "Our - w/o $order_2$," respectively. Their performance also drops significantly. In contrast, our proposed grouping method achieves the best performance in both spatial information reconstruction (PSNR) and spectral structure preservation (SAM).

3) Study of the AE Module: The AE module is designed to learn a correlation representation between the current band and its context information, which can enhance the complementary information of the context at the feature level. Fig. 12 shows some visualization results of the last AE module to demonstrate its effectiveness. Specifically, Fig. 12(b) and (e) shows feature response maps before and after the enhancement of neighborhood information, i.e., $\boldsymbol{N}^t_J$ and $\hat{\boldsymbol{N}}^t_J$. Fig. 12(c) shows the response map of context information $\boldsymbol{C}^t_J$, and (f) displays the output after enhancement ($\hat{\boldsymbol{C}}^t_J$). Correlation maps between the two inputs and the two outputs are calculated, as shown in columns Fig. 12(d) and (g), respectively. Areas with high response values are highlighted in red, indicating

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:29:33 UTC from IEEE Xplore. Restrictions apply.

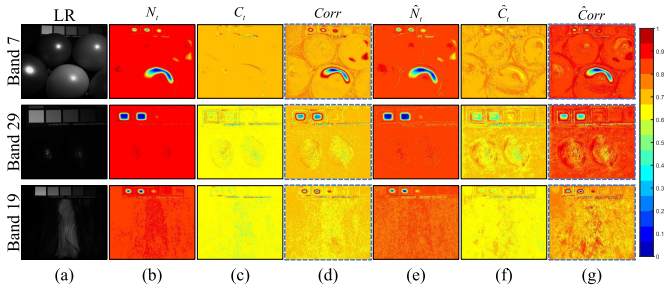


Fig. 12. Visualization of feature maps of the last AE module. (a) LR spectral band; (b) (c) two inputs of the AE module and (d) correlation between them; (e) and (f) two enhanced outputs of the AE module and (g) correlation map between them.

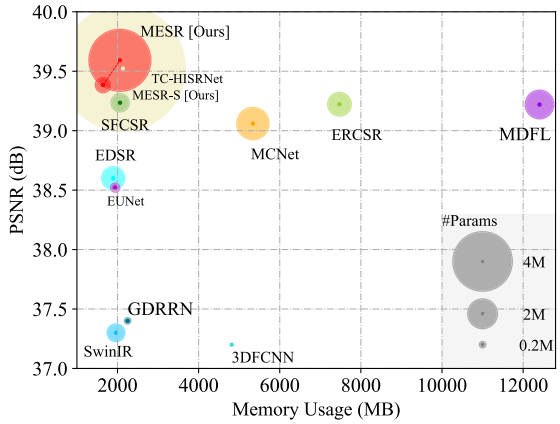


Fig. 13. Efficiency and performance comparison on the CAVE dataset at scale factor 4. The memory usage distributes on the horizontal axis, the horizontal axis represents the accuracy, and the circle area reflects the number of the parameters.

that the network focuses more on them, while areas with low response values are shown in blue.

It is evident that the response maps of the neighbor band information Fig. 12(b) and (e) are generally higher than that of the context information Fig. 12(c) and (f). This phenomenon accords with common sense, as there is a high similarity between neighbor bands, which contributes more to the reconstruction result. Moreover, columns Fig. 12(d) and (g) demonstrate a higher correlation between the context and neighbor band information after passing through the AE module. In addition, by comparing Fig. 12(c) with (f), an overall increase in the response values of the context can be observed. This further indicates that the proposed AE module can effectively enhance complementary information in the context so that the network pays more attention to the content introduction of the context.

4) Efficiency and Performance Comparison: By comparing the accuracy, memory usage, and number of parameters of each method, we perform the comprehensive advantages of the proposed method in terms of performance and efficiency. As seen in Fig. 13, our method is more accurate than other compared peers (the quantitative results are summarized in Table II). For memory efficiency, although slightly less than some nature image SR methods with poor accuracy performance, i.e., EDSR and SwinIR, our method still has a competitive result. Even including a great number of parameters, benefiting from the recurrent mechanism, the proposed method can still work on some devices with limited memory space. After a synthetic analysis for performance and practicability, it can conclude that our method achieves the best performance overall. In addition, the corresponding efficient version, called MESR-S, contains the fewest parameters and can still achieve optimal performance in both accuracy and efficiency.

5) Ablation Study: The proposed network mainly consists of two parts: the neighbor network and the context network. The neighbor network is composed of four components: 2DBlock, NeiSE, 3DBlock, and a feedback embedding between the neighbor networks (FENei). And the context network is cascades of AE modules. In this section, we perform ablation experiments to analyze the impact of different components on the model performance. The investigation is conducted on the CAVE dataset at scale factor 2.

As shown in Table IX, the results vary with different combinations of modules. Initially, when all the components

TABLE IX
EXPERIMENTAL RESULT OF ABLATION STUDY ON THE CAVE DATASET AT THE SCALE FACTOR 2

Component	Different combinations of modules							
Context feature	✔	✔	✘	✘	✘	✘	✘	✘
Context network	✔	✔	✔	✘	✘	✘	✘	✘
FENei	✔	✔	✔	✔	✘	✘	✘	✘
Neighbor SR	✔	✘	✔	✔	✔	✔	✔	✔
NeiSE	✔	✔	✔	✔	✔	✔	✘	✘
2DBlock	✔	✔	✔	✔	✔	✔	✔	✘
3DBlock	✔	✔	✔	✔	✔	✘	✘	✔
PSNR	45.805	45.753	45.698	45.655	45.533	45.492	42.175	44.241
SSIM	0.9742	0.9741	0.9741	0.9741	0.9740	0.9739	0.9701	0.9731
SAM	2.187	2.192	2.198	2.211	2.216	2.221	2.555	2.262

are attached, the model achieves the best performance on each evaluation metric. First, we discard the loss constraint on the neighbor SR result $\boldsymbol{I}_t^{\text{Nei}}$ and the overall performance is degraded. This is due to the lack of guidance for the neighborhood information modeling, making it difficult to fully exploit information between neighbor bands. Next, the context feature $\boldsymbol{F}_{t-1}^{\text{Con}}$ is removed and replaced by the neighbor feature $\boldsymbol{F}_t^{\text{Nei}}$. And then the context network is further discarded. The results are really poor, especially for the values of PSNR. This indicates that the introduction of context information and the further correlation enhancement for context content are both crucial for improving performance.

Finally, we gradually remove components from the neighbor network, i.e., 2DBlock, NeiFE, 3DBlock, and FENei, to demonstrate their effectiveness. Note that the network still performs well when the parallel mechanism is preserved (NeiFE, 2DBlock). The reason of this phenomenon is that the proposed mechanism enables the network to fully explore the spatial information of the current band and its neighbor ones, while effectively maintaining the spectral structure. Based on the aforementioned analysis, it can be concluded that all the components are effective for network performance improvement.

V. CONCLUSION

In this article, we propose a two-stage network based on neighbor spectra maintenance and context correlation enhancement for single HIS, called MESR. The proposed MESR

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:29:33 UTC from IEEE Xplore. Restrictions apply.

consists of two subnetworks: neighbor network and context network. Specifically, neighbor bands are first divided into several subgroups orderly, and then processed in parallel in the neighbor network. This structure can not only improve the ability to excavate the spatial feature among neighbor bands but also avoid spectral distortion to some extent. Under the action of the neighbor network, the spatial–spectral feature of neighborhoods is generated. As for the context network, a correlation representation between the neighbor feature and the feedback context information is constructed by AE modules, which encourages our network to pay more attention to the contents with higher correlation. Passing through a series of cascading AE modules in the context network, more complementary information in the context is effectively enhanced in this stage. Meanwhile, in addition to the final reconstruction result, we also supervise the super-resolved result generated by neighborhood networks using a loss function, which can improve the stability of intermediate information. The quantitative and qualitative experiments on nature and remote sensing hyperspectral image datasets demonstrate the superiority of our methods in reconstruction accuracy and memory occupation. It means that the proposed network can operate on some memory limited devices, improving its practical applicability.

In the future, it is necessary to simplify the network while ensuring high performance. Although the proposed methods have achieved outstanding performance with efficient memory utilization, the model still exhibits a large number of parameters and high FLOPs values. In addition, improving the performance of SR models in real-world scenarios remains an area that warrants further exploration.

REFERENCES

[1] D. Landgrebe, "Hyperspectral image data analysis," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 17–28, Jan. 2002.

[2] A. Marinoni and H. Clenet, "Identification of mafic minerals on Mars by nonlinear hyperspectral unmixing," in *Proc. 8th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Aug. 2016, pp. 1–4.

[3] Y. Liu and Y. Zhang, "Hyperspectral image refined plant classification by graph based composite kernel," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 7444–7447.

[4] Y. Tan, L. Lu, L. Bruzzone, R. Guan, Z. Chang, and C. Yang, "Hyperspectral band selection for lithologic discrimination and geological mapping," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 471–486, 2020.

[5] A. C. Karaca, D. Cesmeci, A. Erturk, M. K. Gullu, and S. Erturk, "Hyperspectral change detection with stereo disparity information enhancement," in *Proc. 6th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Jun. 2014, pp. 1–4.

[6] G. Zhang, M. Xu, Y. Zhang, and Y. Fan, "Improved hyperspectral anomaly target detection method based on mean value adjustment," in *Proc. 10th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Sep. 2019, pp. 1–4.

[7] M. Ning, P. Yu, W. Shaojun, and G. Wei, "A weight SAE based hyperspectral image anomaly targets detection," in *Proc. 13th IEEE Int. Conf. Electron. Meas. Instrum. (ICEMI)*, Oct. 2017, pp. 511–515.

[8] M. E. Paoletti, J. M. Haut, R. Fernandez-Beltran, J. Plaza, A. J. Plaza, and F. Pla, "Deep pyramidal residual networks for spectral–spatial hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 740–754, Feb. 2019.

[9] N. He et al., "Feature extraction with multiscale covariance maps for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 755–769, Feb. 2019.

[10] R. C. Patel and M. V. Joshi, "Super-resolution of hyperspectral images: Use of optimum wavelet filter coefficients and sparsity regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 4, pp. 1728–1736, Apr. 2015.

[11] C. Wang, W. Pedrycz, M. Zhou, and Z. Li, "Sparse regularization-based fuzzy C-means clustering incorporating morphological grayscale reconstruction and wavelet frames," *IEEE Trans. Fuzzy Syst.*, vol. 29, no. 7, pp. 1826–1840, Jul. 2021.

[12] C. Wang, W. Pedrycz, J. Yang, M. C. Zhou, and Z. Li, "Wavelet frame-based fuzzy C-means clustering for segmenting images on graphs," *IEEE Trans. Cybern.*, vol. 50, no. 9, pp. 3938–3949, Sep. 2020.

[13] C. Wang et al., "Residual-driven fuzzy C-means clustering for image segmentation," *IEEE/CAA J. Autom. Sinica*, vol. 8, no. 4, pp. 876–889, Apr. 2021.

[14] W. Xie, X. Jia, Y. Li, and J. Lei, "Hyperspectral image super-resolution using deep feature matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 6055–6067, Aug. 2019.

[15] C. Wang, Z. Y. Yan, W. Pedrycz, M. C. Zhou, and Z. W. Li, "A weighted fidelity and regularization-based method for mixed or unknown noise removal from images on graphs," *IEEE Trans. Image Process.*, vol. 29, pp. 5229–5243, 2020.

[16] C. Wang, M. Zhou, W. Pedrycz, and Z. Li, "Comparative study on noise-estimation-based fuzzy C-means clustering for image segmentation," *IEEE Trans. Cybern.*, vol. 54, no. 1, pp. 241–253, Jan. 2024.

[17] C. Wang, W. Pedrycz, Z. Li, M. C. Zhou, and S. S. Ge, "G-image segmentation: Similarity-preserving fuzzy C-means with spatial information constraint in wavelet space," *IEEE Trans. Fuzzy Syst.*, vol. 29, no. 12, pp. 3887–3898, Dec. 2021.

[18] Z. Pan et al., "Super-resolution based on compressive sensing and structural self-similarity for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 9, pp. 4864–4876, Sep. 2013.

[19] J. Hu, Y. Li, and W. Xie, "Hyperspectral image super-resolution by spectral difference learning and spatial error correction," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1825–1829, Oct. 2017.

[20] Y. Li, J. Hu, X. Zhao, W. Xie, and J. Li, "Hyperspectral image super-resolution using deep convolutional neural network," *Neurocomputing*, vol. 266, pp. 29–41, Nov. 2017.

[21] J. Li, R. Cui, B. Li, Y. Li, S. Mei, and Q. Du, "Dual 1D-2D spatial–spectral CNN for hyperspectral image super-resolution," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Oct. 2019, pp. 3113–3116.

[22] Q. Li, Q. Wang, and X. Li, "Mixed 2D/3D convolutional network for hyperspectral image super-resolution," *Remote Sens.*, vol. 12, no. 10, p. 1660, May 2020.

[23] Q. Wang, Q. Li, and X. Li, "Spatial–spectral residual network for hyperspectral image super-resolution," 2020, *arXiv:2001.04609*.

[24] Q. Li, Q. Wang, and X. Li, "Exploring the relationship between 2D/3D convolution for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8693–8703, Oct. 2021.

[25] Q. Li, Y. Yuan, and Q. Wang, "Hyperspectral image super-resolution via multi-domain feature learning," *Neurocomputing*, vol. 472, pp. 85–94, Feb. 2022.

[26] H. Wang, C. Wang, and Y. Yuan, "Asymmetric dual-direction quasi-recursive network for single hyperspectral image super-resolution," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 11, pp. 6331–6346, Nov. 2023.

[27] X. Wang, J. Ma, and J. Jiang, "Hyperspectral image super-resolution via recurrent feedback embedding and spatial–spectral consistency regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503113.

[28] Q. Wang, Q. Li, and X. Li, "Hyperspectral image superresolution using spectrum and feature context," *IEEE Trans. Ind. Electron.*, vol. 68, no. 11, pp. 11276–11285, Nov. 2021.

[29] Y. Fu, Z. Liang, and S. You, "Bidirectional 3D quasi-recurrent neural network for hyperspectral image super-resolution," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2674–2688, 2021.

[30] S. Mei, X. Yuan, J. Ji, Y. Zhang, S. Wan, and Q. Du, "Hyperspectral image spatial super-resolution via 3D full convolutional neural network," *Remote Sens.*, vol. 9, no. 11, p. 1139, Nov. 2017.

[31] C. Dong, C. C. Loy, K. He, and X. Tang, "Learning a deep convolutional network for image super-resolution," in *Proc. 13th Eur. Conf. Comput. Vis.*, Sep. 2014, pp. 184–199.

[32] J. Kim, J. K. Lee, and K. M. Lee, "Accurate image super-resolution using very deep convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 1646–1654.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:29:33 UTC from IEEE Xplore. Restrictions apply.

[33] T. Tong, G. Li, X. Liu, and Q. Gao, "Image super-resolution using dense skip connections," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4799–4807.

[34] B. Lim, S. Son, H. Kim, S. Nah, and K. M. Lee, "Enhanced deep residual networks for single image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 136–144.

[35] J. Kim, J. K. Lee, and K. M. Lee, "Deeply-recursive convolutional network for image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 1637–1645.

[36] Y. Tai, J. Yang, and X. Liu, "Image super-resolution via deep recursive residual network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2790–2798.

[37] Y. Qiu, R. Wang, D. Tao, J. Cheng, and J. Cheng, "Embedded block residual network: A recursive restoration model for single-image super-resolution," in *Proc. IEEE Int. Conf. Comput. Vis.*, Jun. 2017, pp. 4180–4189.

[38] K. Jiang, Z. Wang, P. Yi, and J. Jiang, "Hierarchical dense recursive network for image super-resolution," *Pattern Recognit.*, vol. 107, Nov. 2020, Art. no. 107475.

[39] K. Jiang, Z. Wang, P. Yi, J. Jiang, J. Xiao, and Y. Yao, "Deep distillation recursive network for remote sensing imagery super-resolution," *Remote Sens.*, vol. 10, no. 11, p. 1700, Oct. 2018.

[40] C. Ledig et al., "Photo-realistic single image super-resolution using a generative adversarial network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2017, pp. 105–114.

[41] J. Liang, J. Cao, G. Sun, K. Zhang, L. Van Gool, and R. Timofte, "SwinIR: Image restoration using Swin transformer," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops*, Oct. 2021, pp. 1833–1844.

[42] Y. Cao, J. Xu, S. Lin, F. Wei, and H. Hu, "Global context networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 6, pp. 6881–6895, Jun. 2020.

[43] Z. Lu, J. Li, H. Liu, C. Huang, L. Zhang, and T. Zeng, "Transformer for single image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 457–466.

[44] T. Dai, J. Cai, Y. Zhang, S. -T. Xia, and L. Zhang, "Second-order attention network for single image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 11065–11074.

[45] Y. Xiao, Q. Yuan, K. Jiang, J. He, Y. Wang, and L. Zhang, "From degrade to upgrade: Learning a self-supervised degradation guided adaptive network for blind remote sensing image super-resolution," *Inf. Fusion*, vol. 96, pp. 297–311, Aug. 2023.

[46] W.-S. Lai, J.-B. Huang, N. Ahuja, and M.-H. Yang, "Deep Laplacian pyramid networks for fast and accurate super-resolution," in *Proc. CVPR*, 2017, pp. 5835–5843.

[47] J. Johnson, A. Alahi, and L. Fei-Fei, "Perceptual losses for real-time style transfer and super-resolution," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 694–711.

[48] R. Ran, L.-J. Deng, T.-X. Jiang, J.-F. Hu, J. Chanussot, and G. Vivone, "GuidedNet: A general CNN fusion framework via high-resolution guidance for hyperspectral image super-resolution," *IEEE Trans. Cybern.*, vol. 53, no. 7, pp. 4148–4161, Jul. 2023.

[49] Q. Ma, J. Jiang, X. Liu, and J. Ma, "Learning a 3D-CNN and transformer prior for hyperspectral image super-resolution," *Inf. Fusion*, vol. 100, Dec. 2023, Art. no. 101907.

[50] Y. Yuan, X. Zheng, and X. Lu, "Hyperspectral image superresolution by transfer learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 5, pp. 1963–1974, May 2017.

[51] Z. He and L. Liu, "Hyperspectral image super-resolution inspired by deep Laplacian pyramid network," *Remote Sens.*, vol. 10, no. 12, p. 1939, Dec. 2018.

[52] Y. Li, L. Zhang, C. Dingl, W. Wei, and Y. Zhang, "Single hyperspectral image super-resolution with grouped deep recursive residual network," in *Proc. IEEE 4th Int. Conf. Multimedia Big Data (BigMM)*, Sep. 2018, pp. 1–4.

[53] D. Liu et al., "An efficient unfolding network with disentangled spatial–spectral representation for hyperspectral image super-resolution," *Inf. Fusion*, vol. 94, pp. 92–111, Jun. 2023.

[54] L. Yang et al., "TC-HISRNet: Hyperspectral image super-resolution network based on contextual band joint transformer and CNN," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 9632–9645, 2023.

[55] M. Haris, G. Shakhnarovich, and N. Ukita, "Recurrent back-projection network for video super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3892–3901.

[56] Y. Xiao, X. Su, Q. Yuan, D. Liu, H. Shen, and L. Zhang, "Satellite video super-resolution via multiscale deformable convolution alignment and temporal grouping projection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5610819.

[57] T. Isobe et al., "Video super-resolution with temporal group attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 8005–8014.

[58] M. Lu and P. Zhang, "Grouped spatio-temporal alignment network for video super-resolution," *IEEE Signal Process. Lett.*, vol. 29, pp. 2193–2197, 2022.

[59] D. Liu, J. Li, and Q. Yuan, "A spectral grouping and attention-driven residual dense network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7711–7725, Sep. 2021.

[60] X. Chen et al., "Bi-directional cross-modality feature propagation with separation-and-aggregation gate for RGB-D semantic segmentation," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 561–577.



Heng Wang received the M.E. degree in electronics and communication engineering from the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China, in 2021, where he is currently pursuing the Ph.D. degree with the School of Computer Science and School of Artificial Intelligence, OPtics and Electronics (iOPEN).

His research interests include computer vision and machine learning.



Cong Wang (Member, IEEE) received the B.S. degree in automation and the M.S. degree in mathematics from Hohai University, Nanjing, China, in 2014 and 2017, respectively, and the Ph.D. degree in mechatronic engineering from Xidian University, Xi'an, China, in 2021.

He was a Visiting Ph.D. Student with the Department of Electrical and Computer Engineering, University of Alberta, Edmonton, AB, Canada, and the Department of Electrical and Computer Engineering, National University of Singapore, Singapore. He is now an Associate Professor with the School of Artificial Intelligence, OPtics and ElectroNics (iOPEN), Northwestern Polytechnical University, Xi'an. His research interests include wavelet analysis and its applications, granular computing, and image processing.



Yuan Yuan (Senior Member, IEEE) is currently a Full Professor with the School of Artificial Intelligence, OPtics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an, China. She has authored or coauthored more than 150 articles, including about 100 in reputable journals, such as the IEEE TRANSACTIONS AND PATTERN RECOGNITION, as well as the conference papers in CVPR, BMVC, ICIP, and ICASSP. Her research interests include visual information processing and image/video content analysis.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:29:33 UTC from IEEE Xplore. Restrictions apply.